```
EXEC PARAMETERS 6/21/66  (PARAMS,31)

SZS=640000
LAT=762200

GLPHNG=670000
TYIHNG=700000
TYOHNG=710000
RPAHNG=720000
PPAHNG=730000
CLKHNG=740000
EOTHNG=750000
UNUSED=760000
CORHNG=770000

BITS=0
NOCHAR=1
TOTRT=2
IOPRT=4
BILLTT=6
SUPTD=7
PROGNM=11
PRIORITY=15
DDTSEG=16
RESTPC=32
HH16AC=40
HH16PC=41
HH16IO=42
HH16FG=43
C16AC=44
C16PC=45
C16IO=46
C16FLA=47
AC=50
PC=35
IO=51
FLAGS=52
FASTAC=53
FASTPC=54
C16TEM=55
DDTS6=65
RESTSU=73
IOPMAX=74
TISMAX=75
TTNO=76
TRAPPC=77
ERCODE=102
TTTSU=105
```

```
LRG=IOT 10
ERG=IOT 11
RRC=IOT 2122          RCC=IOT 3022
TCB=IOT 4022          SSB=IOT 4122
RCK=IOT 32
CKS=IOT 33
RTB=IOT 35
DSC=IOT 50
ASC=IOT 51
ISB=IOT 52
CAC=IOT 53
LSM=IOT 54
ESM=IOT 55
CBS=IOT 56
DIA=IOT 61
DWC=IOT 62            DRA=IOT 2062
DCL=IOT 63
ERM=IOT 65
RNM=IOT 66
LEM=IOT 74            EEM=IOT 4074
RPA=IOT 1  RRB=IOT 30
PPA=IOT 5

PUNLEN=40
RDLEN=200
COR=140000

START
```

SEQUENCE BREAK ROUTINES 10/12/66 (SBOUT, 41)

```
0/          OFFSET DCH
7/          JMP DATACH  /1   DATA CHANNEL
13/         JMP READER  /2   READER
23/         JMP CONTRL  /4   CONTROLLERS
27/         JMP LD      /5   SWAPPING DRUM
33/         JMP TTSERV  /6   SCANNER
37/         JMP 1SEC    /7   1 SEC CLOCK
43/         JMP 1MIN    /10  1 MIN CLOCK
53/         JMP PUNCH   /12  PUNCH
63/         JMP SOROBA  /14  TYPEWRITER
67/         JMP IOP     /15  IO PROCESSOR ISB'S
73/         JMP DISPAT  /16  RESTRICT MODE TRAP
77/         JMP SWAP    /17  32 MSEC. CLOCK

200/STAT, TYIHNG        /STAT+177 MUST BE LESS THAN 8 BITS (377)
         REPEAT 177, UNUSED
```

```
/TELETYPE TRANSLATION TABLES
600/      /TABLES START AT ODD MULTIPLE OF 200

/BIT 0 INDICATES 12 BIT CHAR. ON INPUT
/BITS 1-6 ARE INTERNAL CODE OF INPUT CHAR.
/BIT 7 INDICATES CHAR. IS CT ALARM CHAR.
/BIT 8 INDICATES CHAR. IS NON-CT ALARM

/BITS 10-17 ARE TT OUTPUT CODE

W=402000  /WARNING ALARM CHARACTERS
C=002000  /6 BIT CT ALARM CHARACTERS
L=000000  /NON ALARM CHARACTERS
U=000177  /OUTPUT CHARACTER FOR UNUSED INTERNAL CODES

DEFINE M TYPE,IN,OUT
IN"T"4000+TYPE+OUT
TERMINATE M

TTTBL,                      /INPUT CHAR    /OUTPUT CHAR
M L,00,040                  /NULL          /SPACE
M W,01,041                  /"A"           /!
M W,02,042                  /"B"           /"
M W,03,043                  /"C"           /#
M W,04,044                  /EOT           /$
M W,05,045                  /WRU           /%
M W,06,046                  /RU            /&
M W,07,047                  /BELL          /'

M W,10,050                  /"H"           /(
M W,11,051                  /TAB           /)
M W,12,052                  /LF            /*
M W,13,053                  /VT            /+
M W,14,054                  /FORM          /,
M C,76,055                  /CR            /-
M W,16,056                  /"N"           /.
M W,17,057                  /"O"           //

M W,20,060                  /"P"           /0
M W,21,061                  /"Q"           /1
M W,22,062                  /TAPE          /2
M W,23,063                  /XOFF          /3
M W,24,064                  /"T"           /4
M W,25,065                  /"U"           /5
M W,26,066                  /"V"           /6
M W,27,067                  /"W"           /7

M W,30,070                  /"X"           /8
M W,31,071                  /"Y"           /9
M W,32,072                  /"Z"           /:
363073                      /"["=AM        /;
M W,34,074                  /"BS"          /<
M W,35,075                  /"]"           /=
M W,36,076                  /"↑"           />
M W,37,077                  /"←"           /?
```

```
M C,00,100          /SPACE          /@
M C,01,101          /! /A
M L,02,102          /" /B
M C,03,103          /# /C
M C,04,104          /$ /D
M C,05,105          /% /E
M C,06,106          /& /F
M C,07,107          /' /G

M C,10,110          /( /H
M C,11,111          /) /I
M C,12,112          /* /J
M C,13,113          /+ /K
M C,14,114          /, /L
M C,15,115          /- /M
M L,16,116          /. /N
M C,17,117          // /O

M L,20,120          /0 /P
M L,21,121          /1 /Q
M L,22,122          /2 /R
M L,23,123          /3 /S
M L,24,124          /4 /T
M L,25,125          /5 /U
M L,26,126          /6 /V
M L,27,127          /7 /W

M L,30,130          /8 /X
M L,31,131          /9 /Y
M C,32,132          /: /Z
M C,33,133          /; /[
M C,34,175          /< /AM
M C,35,135          /= /]
M C,36,015          /> /CRLF
M C,37,  0          /?
```

```
M C,40,000               /@ /NULL
M L,41,001               /A /"A"
M L,42,002               /B /"B"
M L,43,003               /C /"C"
M L,44,004               /D /EOT
M L,45,005               /E /WRU
M L,46,006               /F /RU
M L,47,007               /G /BELL

M L,50,010               /H /"H"
M L,51,011               /I /TAB
M L,52,012               /J /LF
M L,53,013               /K /VT
M L,54,014               /L /FORM
M L,55,015               /M /CR
M L,56,016               /N /"N"
M L,57,017               /O /"O"

M L,60,020               /P /"P"
M L,61,021               /Q /"Q"
M L,62,022               /R /TAPE
M L,63,023               /S /XOFF
M L,64,024               /T /"T"
M L,65,025               /U /"U"
M L,66,026               /V /"V"
M L,67,027               /W /"W"

M L,70,030               /X /"X"
M L,71,031               /Y /"Y"
M L,72,032               /Z /"Z"
M C,73,033               /[ /"["
M W,44,034               /BS            /"BS"
M C,75,035               /] /"]"
M W,46,036               /↑ /"↑"
M W,47,037               /← /"←"

REPEAT 4, M L,0,U        /  /
M L, 0,134               /  /BS
M L, 0, U /              /
M L, 0,136               /  /↑
M L, 0,137               /  /←

REPEAT 20, M L,0,U       /  /

REPEAT 4, M L,0,U        /  /
M L, 0,177               /  /RO
363000+U                 /EOM           /
363000+U                 /EOM           /
763000+U                 /RO            /

TTTBL+200,               EXPUNGE C,L,M,W,U
```

```
1000,TTP, /TELETYPE PTRS. (400 WDS)

DEFINE TTPTRS NUM,ORG
REPEAT NUM,[
101         /LINE NOT OPEN, RNG SET
LAC 140000+[.-TTP-1]"T"2+ORG        /SERVICE PTR
LAC 140000+[.-TTP-2]"T"2+ORG        /USER PTR
0           /PGM WORD (0=DISCONNECTED)
]
TERMINATE TTPTRS

TTPTRS 10, 100          /100-177 IS TT BUFFERS 0-7
TTPTRS 20, 300          /400-577 IS TT BUFFERS 10-27
TTPTRS 50, 1100         /1400-2077 IS TT BUFFERS 30-77
TTP/        105
TTP+3/      STAT 0

/INITIALIZER
2100/
INIT,       LAC .+2     /FLY THE SCANNER
            DAC 74
            CKS
            RIR 2S
            SPI
            RCC
            ISP 74
            JMP INIT+2
            RTB
            CAC
            CBS
            LAW DSWAP
            DAP 77
            ESM
            ISB 1700
            JMP .
DSWAP,      LAW SWAP
            DAP 77
            LIO (340000)   /MEMORY PROTECT BITS
            ERM
            IRP [A,,1,2,4,5,6,7,12,14]
            ASC A"T"100
            ENDIRP
            ISB 600     /JUST TO MAKE SURE SCANNER FLYING
           EEM
            JMP I .+1
DSWAPX:     DCH ESWORG  /DAC'ED INTO BY RESTART TAPE
```

```
/DATA CHAN SEQ BRK
DATACH,   LIF"U"SCF"U"CLL
          DIO DATSAV
          EEM
          JMP I DATADR
DATX,     LIO DATSAV
          LFI
          LIO 6
          LAC 4
          JMP I 5
DATSAV,   0

/CONTROLLERS SEQ BRK
CONTRL,   LIO CONSAV
          LIF"U"LFI
          DIO CONSAV
          EEM
          JMP I CONADR
CONX,     LIO CONSAV
          LIF"U"LFI
          DIO CONSAV
          LAC 20
          LIO 22
          JMP I 21
CONSAV,   0

/IO PROCESSOR LOW PRIORITY SEQ BRK
IOP,      LIF"U"SCF"U"CLL
          DIO IOPSAV
          EEM
          JMP I IOPADR
IOPX,     LIO IOPSAV
          LFI
          LIO 66
          LAC 64
          JMP I 65
IOPSAV,   0
/READER SEQ BRK

DIMENSION RDRBUF(RDLEN)

READER,   EEM
          JSP RDR
          LIO 12
          LAC 10
          JMP I 11
```

```
RDR,        RRB
RB1,        XX              / RPA OR DIO RB1
            DAC READEX
            LAC RBI
            SPA
            JMP RB3
RB4,        RIL 9S
            DIO I RBI
            ADD (400000)
            SAD RBE
            SUB (RDLEN)
            DAC RBI
            ADD (1)
            SUB RBE
            RAL 1S
            SMA
            SUB (RDLEN RDLEN)
            RAR 1S
            ADD RBE
            SUB RBO
            RAL 1S
            SPA
            ADD (RDLEN RDLEN)
            SZA I
            JMP RB6
            SAS (RDLEN RDLEN-40)
            JMP RB5
7RDRI,      LAC RDWHO
            SAD BDSTAT
            LAW FSTAT
            DAP RDVAR
            LAW I 7777  /BUFFER FILLING
            LSM
            AND I RDVAR
            SAS (RPAHNG)   /HUNG ?
            JMP RB7       /NO
            LIO (3)
            LAW STAT
            SAD RDWHO
            CLI
            DIO I RDVAR
            ESM
            DZM HOTFLG
RB5,        RCK
            DIO RBK
            JMP I READEX
RB3,        LAC I RBI
            RCL 9S
            LAC RBI
            JMP RB4
RB6,        LIO .+1       /BUFFER FULL ON NEXT CHARACTER
CRB1,       DIO RB1
            JMP RB5
RB7,        ESM
            JMP RB5
```

```
RDVAR,    DCH .
RBI,      COR RDRBUF
RBO,      COR RDRBUF
RBE,      COR RDRBUF+RDLEN
RBK,      0
RDWHO,    -0
RX,       0
READEX,   0
```

```
/PUNCH SEQ BRK

DIMENSION PUNBUF(PUNLEN)

PUNCH,    EEM
          IDX PRUN
          LAC PSP
          SAD PUP
          JMP PUNX
          DZM PRUN
          LIO I PSP
          PPA
          IDX PSP
          SAD PEND
          SUB (PUNLEN)
          DAC PSP
          ADD (10)
          SAD PUP
          JMP PUNALM
          SUB (PUNLEN)
          SAS PUP       /10 CHARS FROM FULL?
         JMP PUNX
PUNALM,   LAC PUNWHO
          SAD BDSTAT
          LAW FSTAT
          DAP PVAR
          LAW I 7777
          LSM
          AND I PVAR
          SAS (PPAHNG)   /HUNG ?
          JMP PUNCH1   /NO
          LAW 3
          DAC I PVAR
          ESM
          ISB 1700
         DZM HOTFLG
PUNX,     LIO 52
          LAC 50
          JMP I 51

PUNCH1,   ESM
          JMP PUNX

PUNWHO,   -0
PSP,      COR PUNBUF
PUP,      COR PUNBUF
PEND,     COR PUNBUF+PUNLEN
PVAR,     DCH .
PRUN,     1
```

```
/SOROBAN SEQ BRK
SOROBA,    EEM
           LSM
           JSP I SORADR
SORX,      ESM
           LIO 62
           LAC 60
           JMP I 61
```

```
/1 SEC CLOCK
1SEC,       LIO (1)     /CHANNEL 7 ROUTINE TO UNHANG
           LAW STAT    /USERS WHO ARE CLOCK HUNG
1SECL1,     DAP 1PTR
1SECL2,     LAW I 7777  /LOOP TO SEARCH FOR CLOCK-HUNG
            LSM
1PTR,       AND .
            SAD (CLKHNG)
            JMP 1SECT2  /FOUND ONE
1SECT1,     ESM
            IDX 1PTR
            SAD (AND FSTAT+1)
            JMP 1SECC
            ADD (WHERE-STAT-AND)
            SAS WTOP
            JMP 1SECL2
            LAW FSTAT
            JMP 1SECL1

1SECT2,     IDX I 1PTR  /COUNT TOWARD 7777
            SAS (CLKHNG+I) /IS IT UNHUNG
           JMP 1SECT1
            DIO I 1PTR
            DZM HOTFLG
            JMP 1SECT1

1SECC,      EEM
            LSM
            LAC OCTR
            DZM OCTR
            ESM
            SZA
            JSP I 1SECJA
1SECA,      LSM
            LAC 1CTR
            DZM 1CTR
            ESM
            SZA
            JSP I 1SECJB
1SECB,      SZS I 40
7TT2,       JSP .+1     /NORMAL .+1; ABNORMAL 7TT4. SET ON CH6
            LAC RB1
            SAS RB2
            JMP 7RDRX
            LSM
            RCK
            LAI
            SUB RBK
            ESM
            SPA
            ADD (60000.)
            SUB (100.)
            SPA
            JMP 7RDRX
            LIO .+1
            DIO RB1
            DSC 200
            LAC (DCH 7RDRXX)
            DAC READEX
            JMP 7RDRI
7RDRXX,     ASC 200
7RDRX,      LIO 36
            LAC 34
            JMP I 35
```

0 CTR, 0
1 CTR, 0

```
/1 MINUTE CLOCK
1MIN,   EEM
        JMP I 1MIADR
1MIX,   LIO 42
        LAW 1000
        DAP TTF      /SEE SWNN DISPLAY HACK
        LAC 40
        JMP I 41

START
```

TELETYPE SERVICE ROUTINE 8/15/66 (TTSERV,32)

```
TTSERV,   LIF
          DIO TTFLGS
TTLOOP,   ASC 600      /PROCESS ALL WAITING CHARS, THEN DEBREAK
         CKS
          RIR 2S
          SPI I
          JMP TTDBRK   /SCANNER NOT STOPPED
         RRC
          SZS 40
          JMP TTSS4    /IGNORE HIGH-NUMBERED TT'S
TTSS4X,   SSB
          LAW TTP"T"200000
          SCF"U"AAI
          SAL 2S
          DAP . 1
TTF,      LIO .
          LFI"U"IDA
          DSC 600      /KEEP WAITING BKS FROM PILING UP
         RCC
          SZF I 6
          JMP TTLO     /LINE OPEN
          SZF I 1
          JMP TTIN     /INPUT
          DAP TTALD    /OUTPUT
          DAP TTASP
          IDA
          DAP TTAUP
          LAW TTTBL    /ODD INTEGRAL MULTIPLE OF 200
          IAI"U"SZL"U"CLL
          JMP TTASP    /DON'T TEST FOR NULL OR CR
          SZF 3
          JMP TTINT    /BEING INTERRUPTED
          SAD (TTTBL
          JMP TT1NUL   /FIRST NULL
          SAD (TTTBL"U"215
          JMP TTACR    /CR.  SEND A LF.
TTASP,    LAC .
TTAUP,    SAD .
          JMP TTEMTY   /GO PASSIVE
          IDA"U"IDC
          SAD I TTAUP
          CLF 6        /PF 6 TEM STORAGE FOR ALARM CONDITION
TTALD,    LCH I .
          SAD (770000
          JMP TTA77    /TWO-BYTE CHAR
          DCH (JMP .+1
          LIO TTTBL+0 /GET OUTPUT CODE
TTFULX,   TCB
          SZF I 6
          JMP TTALM    /UNHANG IF HUNG
          LIF
          LAI
          DAP I TTF
          JMP TTLOOP
```

```
/ROUTINE FOR INPUT

TTIN,     DAP TTIDCH
          SZF 2
          JMP TT8BIT
          DAP TTISP
          IDA"U"SZL"U"CLL
          JMP TTIIGN  /IGNORE.  GENERATED BY SERVICE ROUTINE.
         DAP TTIUP
TT8INT,   LAW TTTBL   /ODD INTEGRAL MULTIPLE OF 200
         IAI
          SZF 3
          JMP TTINT   /BEING INTERRUPTED
          SAD (TTTBL
          JMP TT1NUL  /FIRST NULL
          SZF 5
          JMP TTFULL  /NO ROOM IN BUFFER
         DAP . 1
          LIO .       /GET WORD FROM TRANSLATION TABLE
TTISP,    LAC .
          IDC
TTIUP,    SAD .
          JMP TTFUL1
          SPI
          JMP TTI77
          IDA"U"IDC
          SAD I TTIUP
          CLF 6       /PF 6 TEM STORAGE FOR ALARM CONDITION
TFU1X,   LAI
          SAD TTTBL+15
          JMP TTINCR  /CR INPUT
TTINCX,   RAL 1S
TTIDCH,   DCH I .     /PUT CHAR IN BUFFER
          SPA"U"SZF 4
          JMP TTALM   /CONTROL MODE ALARM
          RAL 1S
          SMA"U"SZF 6 I
          JMP TTALM   /REGULAR ALARM OR BUFFER NEARLY FULL
TTIIGN,   LIF
          LAI
          DAP I TTF
          JMP TTLOOP
```

```
/BRANCHES AND SPECIAL CASES

/SS 4 UP
TTSS4,     LAW I 1
           SCF"U"AAI
           SPA
           JMP TTSS4X  /LOW NUMBER
           LAW 7TT4    /SET SWITCH IN CH 7 ROUTINE
          DAP 7TT2
           RCC         /FLUSH.  NO TIME TO CKS, SO DEBREAK.
TTDBRK,    LIO TTFLGS
           LFI
           LAC 30
           LIO 32
           JMP I 31

/ROUTINE JSP'ED TO ON CH 7 BREAK
/SEND RUBOUT ON ALL TT'S TYPEACTIVE WITH LINE CLOSED
7TT4,     DAP 7TT2
           LAW 3       /LOWEST TT IGNORED WHEN SS4 IS UP
          DAC 7TTEM
           LSM
7TT5,      LIA
           SSB
           ADD (TTP"T"200000
           SAL 2S
           DAP . 2
           LAW 41
           AND .
           CLI"U"CMI   /RUBOUT IS 377
           SAD (41     /TYPEACTIVE AND LINE CLOSED
          TCB
           IDX 7TTEM
           SAS (100
           JMP 7TT5
           ESM
           JMP 7TT2+1

7TTEM,     0

/LINE OPEN
TTLO,      LAW TTTBL
           SZL"U"CLL"U"CML"U"IAI
           JMP TTLGC   /FINISH CLOSING LINE
           DAP . 2
           LAW I 3777
           AND .       /TRANSLATE CHAR
           SAS (360000 /EOM?
           JMP TTLOOP  /NO.  IGNORE.
           LIF"U"IAI   /ADR PART OF CLEAR
           DAP I TTF
TTINT1,    LIO (215)   /START CLOSING LINE OR RUBOUT ECHOED
           TCB
           JMP TTLOOP
```

```
/FINISH CLOSING LINE
TTLGC,    LAW 212     /LF
          CLI"U"SWP"U"STF 6
          TCB
          LIF"U"SCF"U"IAI              /LEAVE RING MODE
          DAP I TTF
          LAC (140000+TTLCTX
          JDA TTSOT


/BEING INTERRUPTED BY NULL.  PF 3 UP.
TTINT,    SAD (TTTBL"U"377
          JMP TTINT1  /RUBOUT ECHO.  SEND CR.
          SAD (TTTBL"U"215
          JMP TTBRK   /CR ECHO.  SEND LF AND GIVE "BREAK".
         LAC (I
          ADD I TTF
          DIP I TTF   /INCREMENT NULL COUNT
          SAS I TTF   /END-AROUND CARRY?
          JMP TTLGO   /YES.  SET LINE TO "OPEN".
          CLI"U"CMI
          TCB
          JMP TTLOOP

/LINE GOING OPEN
TTLGO,    LAW 11
          LIF"U"SCF"U"XAI              /CLF 3 AND CLF 6 AND LRG
         DAP I TTF
          LAC (140000+TTLOTX
          JDA TTSOT

/FIRST NULL
TT1NUL,   CLI"U"CMI"U"STF 3
          TCB         /RUBOUT
          LAC (I      /NULL COUNT SET TO "1"
          LIF"U"IAI
TTSUPX,   DAC I TTF   /SIC.
          JMP TTLOOP
```

```
/GIVE "BREAK" ALARM AND SEND LF
TTBRK,     DZM HOTFLG
           ISB 1700
           LIO (212    /LF
           TCB
           LAW 3
           ADD TTF
           DAP TTBRK1
           LAW I 1000  /ALL BUT "77 SAVED" BIT
TTBRK1,    AND .
           SZF I 1     /FLUSH BUFFER IF OUTPUT
          JMP TTBRKA  /INPUT
           DAC I TTBRK1    /OUTPUT
           LIO I TTAUP
           DIO I TTASP
TTBRKA,    AND (377)   /KEEP STAT POINTER
          SZA I
           JMP TTSUP   /STARTUP PROCEDURE
           SAD BDSTAT
           LAW FSTAT
           DAP TTBRK2
           EEM
           SAD TSTAT   /IS IT TOT
           DZM I TOTFLG
           LEM
           LAC (300000
           LSM
TTBRK2,    AND .
           CLI"U"SWP
           SNI
           LAC I TTBRK2
           IOR (40000  /SET MY BIT
          DIP I TTBRK2
           ESM
           LAW 210
          LIF"U"XAI   /SET LINK AND CLF 3
           DAP I TTF   /LEAVE NON-ZERO NULL COUNT AS "BREAK"
          JMP TTLOOP
```

```
/STARTUP PROCEDURE
TTSUP,   LAW 7774
         AND TTBRK1
         SAR 2S       /GET TT NUMBER+200
         LIA
         LAW WHERE
         DAP TTSUP2
TTSUP1,  IDX TTSUP2   /SEARCH WHERE TABLE FOR 300
         LAW 300
TTSUP2,  SAS .
         JMP TTSUP1
         DIO I TTSUP2   /PUT TT NUMBER+200 IN "WHERE" TABLE
         LAW I WHERE-STAT
         ADD TTSUP2
         DAP I TTBRK1   /SET UP PGM WORD
         DAP TTSUP3
         SUB (SAS STAT-WHERE)
         SAD WTOP
         IDX WTOP
         LAW 1        /GIVE HI QUEUE
TTSUP3,  DAC .
         LAW 210
         LIF"U"XAI    /SET LINK AND CLF 3 AND CLEAR NULL COUNT
         JMP TTSUPX
```

```
/BUFFER ALARM.  UNHANG IF HUNG.
TTALM,     CLI"U"STF 6 /RESTORE PF 6
           LAW 3
           ADD TTF
           DAP . 2
           LAW 377
           AND .
           SZA I
           JMP TTALMX  /HAS NO PGM.  FLUSH ALARM.
          EEM
           SAD TSTAT
           DZM I TOTFLG
           LEM
           SAD BDSTAT
           LAW FSTAT
           DAP TTALM1
           SAS (STAT)  /IS IT EXEC DDT
           LIO (1)
          LAW I 7777
TTALM1,    AND .
           SAS (TYOHNG)
          SAD (TYIHNG)
           JMP TTALM2
           SAS (GLPHNG)
           JMP TTALMX
TTALM2,    DIO I TTALM1
           DZM HOTFLG
TTALMX,    LIF"U"IAI   /ADR PART OF AC CLEAR
           DAP I TTF
           JMP TTLOOP

/TYPEACTIVE CR ECHO.  SEND LF.  DON'T STORE FLAGS.
TTACR,     LIO (212    /LF
           TCB
           JMP TTLOOP

/TYPEACTIVE, GOING PASSIVE: BUFFER EMPTY
TTEMTY,    LAW 40
          LIF"U"XAI   /CLF 1
           DAP I TTF
           JMP TTLOOP

/TYPEACTIVE, 12-BIT CHARACTER PROCESSING
TTA77,     LCH I TTALD
           SAS (150000
           SZA I
           CML         /IGNORE 7715 AND 7700 ECHOS
          DCH (JMP .+1
           LIO TTTBL+100  /GET TT CODE
          TCB
           LAC I TTASP
           IDA"U"LIF
           SZF 6
           SAD I TTAUP
           JMP TTALM   /GIVE ALARM
           LAI
           DAP I TTF
           JMP TTLOOP
```

```
/8-BIT MODE INPUT
TT8BIT,    DAP TT8SP
           IDA"U"SZL"U"CLL
           JMP TTIIGN  /IGNORE.  GENERATED BY SERVICE ROUTINE.
          DAP TTIUP
           IDA
           DAP TT8PGM
           SZF 3
           JMP TT8INT  /BEING UNTERRUPTED
          SNI
           JMP TT1NUL  /FIRST NULL.  ONLY FULL NULLS WORK.
          RIR 8S
           LAW I 1777  /ALARM CHAR IN TOP 8 BITS OF PGM WORD
TT8PGM,    AND .
          XAI
          SZA I       /TEST FOR ALARM CHAR
          CLF 6       /PF 6 TEM STORAGE FOR ALARM CONDITION
TT8SP,    LAC .
          IDC
          SAS I TTIUP
          SZF 5
           JMP TTFULL  /BUFFER TOO FULL
           JMP TT8X

/FULL BUFFER ON INPUT
TTFULL,    CML         /IGNORE ECHO
           LIO (334    /BACKSLASH
           JMP TTFULX  /SEND CHAR, STORE FLAGS.

/ALMOST FULL BUFFER ON INPUT
TTFUL1,    SPI         /TWO-BYTE CHAR?
          JMP TTFULL  /YES.  TOO FULL.
          STF 5       /NO.  ROOM FOR THIS AND NO OTHERS.
          JMP TTFU1X

/12-BIT CHARACTER PROCESSING ON INPUT
TTI77,     SIR 5S
TT8X,      IDC
           SAD I TTIUP
           STF 5       /BUFFER FILLING, WITH PTRS TO BE EQUAL
          IDA
           SAD I TTIUP
           JMP TTI77A  /BUFFER FILLING.  GIVE ALARM.
          IDC
           SAD I TTIUP
TTI77A,    CLF 6       /PF 6 IS TEM STORAGE FOR ALARM
           LAI
           DCH I TTIDCH   /STORE "77"
           JMP TTIDCH

/CR ON INPUT.  SEND LF, TO BE IGNORED ON ECHO.
TTINCR,    CML
           LIO (212
           TCB
           JMP TTINCX
```

```
/TYPE PGM, TT, AND "OPEN" OR "CLOSED" ON SOROBAN
TTSOT,  0               /JDA WITH TEXT PTR
        LAW 3
        ADD TTF
        DAP . 2
        LAW 377         /CALCULATE PGM NUMBER
        AND .
        SUB (STAT
        SAD (-STAT
        LAW 177
        RCR 9S
        LAW I 3
        EEM
        LSM
        JDA 14SOCT      /3 DIGITS AND SPACE
        LAC TTF
        RCR 8S          /TTP IS 1000
        LAW I 2
        JDA 14SOCT      /2 DIGITS AND SPACE
        LAC TTSOT
        JDA 14STXT
        ESM
        LEM
        JMP TTLOOP

TTLOTX, 464765          /"OPEN" WITH CR AND "EOM"
        457756

TTLCTX, 634346          /"CLOSED" WITH CR AND "EOM"
        226564
        775600

/CALL SOROBAN OCTAL PRINT IN CORE 15
14SOCT, 0
        DAP 14SOCX
        LAC 14SOCT
        DAC I 15SOCT
        JSP I 15SOC1
14SOCX, JMP .

/CALL SOROBAN TOS IN CORE 15
14STXT, 0
        DAP 14STXX
        LAC 14STXT
        DAC I 15STXT
        JSP I 15STX1
14STXX, JMP .

TTFLGS, 0

START
```

```
DISPATCHER 10/13/66  =DISPAT,64=

/CHAN 16 BREAK
DISPAT,   LIF"U"SCF"U"CLL
          DIO DSP1      /SAVE FLAGS
          RCL 2S        /TOP OF IO CLEAR
          EEM
          SAD ([DDTS6+3]"T"4)       /IS IDDT "EXECUTING" INSTR?
          JMP DSP3      /YES, GO ADJUST PC FOR CALLING SEQUENCES
DSP4,     RTB
          DIO TRPBUF
          CLA"U"SWP
          SMA           /AN IOT?
          JMP ELGIOT    /NO
          RAL 5S        /YES, TEST I BIT
          SPA
          JMP DSP16     /YES, CORE 16 IOT
          RAL 1S
          DCH (JMP 140000 DSP2)     /REGULAR DISPATCH
         RCL 6S
          LFI
DSP2,     JMP I DSPTB
DSP3,     LAW I 1       /DDT PC ADJUST
          ADD I (DDTS6+4)
          DAP 71
          JMP DSP4

DSP16,    RAL 1S        /CORE 16 DISPATCH
         DCH (JMP 140000 DSP5)
          RCL 6S        /GET REDISPATCH BITS INTO IO
         LFI
          LAC 70        /MOVE AC,PC,IO
          DAC I (C16AC)
          LAC 71
          DAC I (C16PC)
          LAC 72
          DAC I (C16IO)
          LAC DSP1      /FLAGS, TOO
          DAC I (C16FLA)
          LAC I (BITS)
          AND (-40000)     /RESTART MODE DEBREAK LEGAL BIT
         DAC I (BITS)
DSP5,     LAC C16TB     /GET DEBREAK DISPATCH
DSP12,    DAC 71        /DEBREAK, DISPATCH
         JMP I 71
```

```
ILGIOT,   EEM
ELGIOT,   LIO I (BITS)   /ILLEGAL INSTR.
          LAI
          AND (-500000)  /IDDT AND RESTART BITS
          DAC I (BITS)
          SPI I          /IS GUY UNDER IDDT
          JMP DSP8       /NO
          LAC C16BKP     /BREAKPOINT, GET DDT
DSP9,     LIO 71         /SET UP CRASH AC,PC,IO,C16 TEMP STORAGE
          DAC 71
          DIO I (HH16PC)
          JSP C16TSV
          LIO 70
          DIO I (HH16AC)
          LIO 72
          DIO I (HH16IO)
          LIO DSP1
          DIO I (HH16FG)
          LIO TRPBUF
          JMP I 71

DSP8,     RIL 1S
          SPI
          JMP XDDTBK     /HELD BY XDDT + CRASHED
          LAC TRPBUF
          SAS (100400)   /HLT?
          JMP DSP10
          RIL 3S
          SPI
          JMP REALHALT
          LAC (20000) /REALHALT LEGAL BIT
          IOR I (BITS)
          DAC I (BITS)
          LAC C16HLT
          JMP DSP12

DSP10,    LAC C16HH      /HALT HORRIBLE
          JMP DSP9
DSP1,     0
TRPBUF,   0
```

```
/HELD BY XDDT + CRASHED; TYPE ON SOROBAN
XDDTBK,    LAW I STAT
           ADD RSTAT
           RCR 9S
           LAW I 3
           LSM
           JDA 14SOCT
           LAW 77
           AND I (TTNO)
           RCR 6S
           LAW I 2
           JDA 14SOCT
           LAC (DCH XDDTBT)
           JDA 14STXT
           LAC (CORHNG)
           LEM
           DIP I RSTAT
           JMP HANG1E

XDDTBT,    276464
           237756       /XDDT[CR][EOM]

HANG,      RCR 6S       /HANG GUY WITH STATUS IN AC (12-17)
           JSP HANGS    /SUBROUTINE IS IN TT ROUTINES
HANG1,     ISB 1700
           DZM HOTFLG   /TELL SWAPPER
           JMP R0

.RCK,      RCK          /REAL RCK
           JMP DPEEK1

DELAY,     SZF 1
           JMP CLOCK
           ISB 1700
           DZM I (PRIORITY)
DELAY1,    LAW I 1
           ADD RUNTM
           AND RUNTM    /TAKE LOW ORDER BIT FROM RUNTM
          AND (17777)
           SZA I
           JMP R1
          DAC RUNTM
           JMP DELAY1

CLOCK,     IDX 71       /HANG FOR NUMBER OF SECONDS IN AC
          LAW 7777
           SUB 70
           DAP RUNTM
           LAW CLKHNG"T"100
           JMP HANG
```

```
REALHALT, LEM
          LAC (UNUSED)
          DAC I RSTAT
          LAW WHERE-STAT
          ADD RSTAT
          DAP REALH1
          LAW 300
REALH1,   DAC .
          DZM I RCORE
REALH3,   LAW I 1
          LSM
          ADD WTOP
          DAP REALH2
          LIA
          LAW 300
REALH2,   SAS .
          JMP HANG1E
          DIO WTOP
          ESM
          JMP REALH3

HANG1E,   ESM
          JMP HANG1

/DEBREAK FROM CORE 16 THRU HH16
C16RH,    LAW 77
          AND I (BILLTT)
          JDA TTNUM
          LAW C16RH1
          DAP TTCKSX
          JMP TTCKS1+2
C16RH1,   SAS (2)
          JMP C16RH2
          LAC C16BKK
          DAC 71
          JMP R0
C16RH2,   LAC (400000)
          IOR I (BITS)
          DAC I (BITS)
          LIO I (HH16AC)
          DIO 70
          LIO I (HH16PC)
          DIO 71
          LIO I (HH16IO)
          DIO 72
          LIO I (HH16FG)
          JMP R0+1
```

```
/RESTART MODE CONROL IOT
RSMC,     SZF I 5
          JMP RSMC1    /DON'T DEBREAK
          LAC (40000) /DEBREAK LEGAL BIT
         AND I (BITS)
          SZA I
          JMP ELGIOT
          LAW I 1
          ADD I (RESTPC)
          DAC 71
          LIA
          JSP C16TUS  /UNSAVE CORE 16 IF NECESSARY

RSMC1,    LAC I (BITS)
          AND (-140000)  /RESTART MODE AND DEBREAK LEGAL BITS
         SZF 6
          IOR (100000)   /RESTART MODE BIT
          DAC I (BITS)
          JMP R1
```

```
/GET A CHARACTER FROM THE READER BUFFER
/R1=NOT YOURS, R2=EMPTY, R3=OK, CHAR IN IO
RDA,        DAC RX
            LAC RSTAT
            SAS RDWHO
            JMP I RX
            IDX RX
            LAC RB1
            SAD RB2
            JMP R4      /READER RUNNING
            LAC RBI
            SUB RBO
            RAL 1S
            SPA
            ADD (RDLEN RDLEN)
            SUB (20)
            SPA
            JMP R5      / START READER ABOUT 20 CHARACTERS IN BUFF
R4,         LAC RBO
            SAD RBI
            JMP I RX    /EMPTY
            LIO I RBO
            ADD (400000)
            SAD RBE
            SUB (RDLEN)
            DAC RBO
            SPA
            RIR 9S
            RCR 8S
            CLI
            RCL 8S      / IO BITS 10-17
            IDX RX
            JMP I RX
R5,         LAC RB2
            DAC RB1
RB2,        RPA
            RCK
            DIO RBK
            JMP R4

RDRLS,      DAC RX      /RELEASE THE READER
           CLI"U"CMI
            LAC RSTAT
            SAD RDWHO
            DIO RDWHO
            JMP I RX
```

```
/GET READER ROUTINE
XRDIN,    LIF"U"SCF
RDINIT,   DAC RX
          LAC RDWHO
          SAD RSTAT
          JMP .+3
          SAS (-0)
          JMP RDIN1     /R1, OWNED BY ANOTHER USER
         LAC RSTAT
          DAC RDWHO
          SZF 1
          JMP .+4
         LAC (COR RDRBUF)            /GET POINTER TO CORE 15
         DAC RBI
         DAC RBO
          LAC CRB1
          DAC RB1
          IDX RX
RDIN1,    LFI
          JMP I RX
/WRITE P POINTER
DWPP,     LAW I 7       /CHECK TO SEE IF VALID RANGE
         ADD 70
          SZS 50
          JMP DWPP1
          SPA
          JMP ELGIOT
          SUB (40-7)
          SMA
          JMP ELGIOT
DWPP1,    LAC 72
          DAC I 70
          JMP R1

/READ P-POINTER
DPEEK,    LIO I 70
          SZF 1
          LIO I 72
DPEEK1,   SZF 2
          DIO 72
          SZF I 2
          DIO 70
          JMP R1
```

```
/DISPATCH CLEAR THE READER
DCLR,     JSP RDINIT
          JMP R1       /OWNED BY ANOTHER USER
          JMP R2       /OK

/DISPATCHER RPA
DRPA,     JSP RDA
          JMP ELGIOT   /UNOWNED BY USER
          JMP . 3
          DIO 72
          JMP R1       /RETURN TO USER WITH CHAR IN IO
         LAW RPAHNG"T"100
          JMP HANG

/DISPATCHER RELEASE THE READER
DRLSRD,   JSP RDRLS
          JMP R1

/DISPATCHER GET THE PUNCH
DGETPU,   LAC PUNWHO
          SAD RSTAT
          JMP R2
          SAS (-0)
          JMP R1       /OWNED BY ANOTHER USER
          LAC RSTAT
          DAC PUNWHO
          JMP R2

/DISPATCHER RELEASE THE PUNCH
DRLSPU,   CLI"U"CMI
          LAC RSTAT
          SAD PUNWHO
          DIO PUNWHO
          JMP R1

/DISPATCHER PPA
DPPA,     LAC RSTAT
          SAS PUNWHO
          JMP ELGIOT
          LAC PUP
          IDA
          SAD PEND
          SUB (PUNLEN)
          SAD PSP
          JMP PFULL
          LIO 72
          DIO I PUP
          DAC PUP
          LSM
          LAC PRUN
          SZA
          ISB 1200
          ESM
          JMP R1
PFULL,    LAW PPAHNG"T"100
          JMP HANG
```

```
/IOT TO START A USER
PSUC,   ESM
PSU,    LEM
        LAW WHERE-1
        DAP PSUA
        IDX PSUA
        LAW 300
PSUA,   SAS .
        JMP .-3
        LIO (500)
        LSM
        SAS I PSUA
        JMP PSUC
        DIO I PSUA
        LAW 7777
        AND PSUA
        SAD WTOP
        IDX WTOP
        ESM
        LAW I WHERE-STAT
        ADD PSUA
        DAP PSUB
        LAW 1
PSUB,   DAC .
        DZM HOTFLG
        JMP R1
```

```
/DISPATCHER TT ROUTINES

/SET UP PTRS USED BY DISPATCHER TT ROUTINES
TTSET,     DAP TTSETX
           LAW 77
           AND I (TTNO
           JDA TTNUM
TTSETX,    JMP .

TTNUM,     0            /JDA WITH TT NUMBER IN AC
           DAP TTNUMX
           LAW TTP"T"200000
           ADD TTNUM
           SAL 2S       /GET PLACE IN TTP TABLE
           DAP TTFP
           IDA
           DAP TTSP
           IDA
           DAP TTUP
           DAP TTDCHU
           DAP TTLCHU
           IDA
           DAP TTPP
TTNUMX,    JMP .

TTFP,      140000+.     /FLAGS
TTSP,      140000+.     /SERVICE PTR
TTUP,      140000+.     /USER PTR
TTPP,      140000+.     /PROGRAM


/CHECK TT STATUS IOT
TTCKS,     JSP TTCKS1
           SZA I
           JMP TTRETN
           DAC 70       /PUT ERROR CODE IN AC AND ERROR CODE WORD
           JMP TTCKS2   /CLEAR OUT BREAK IF ANY


/SUBROUTINE TO CHECK OWNERSHIP AND BREAK
TTCKS1,    DAP TTCKSX
           JSP TTSET    /TT NUMBER IS IN USER CORE
TTCKS1+2,  LAW 377
           AND I TTPP
           LIO (1
           SAS RSTAT
          JMP TTCKSY   /NOT HIS.  ERROR CODE 1.
          LAW I 7767   /770010 TO AC
           AND I TTFP
           RAR 4S
           RIL 1S
           SPQ          /BREAK?
           CLI          /NO
TTCKSY,    LAI
TTCKSX,    JMP .
```

```
/SUBROUTINE NORMALLY USED BY TT IOT'S.  CHECKS OWNERSHIP
/AND DETECTS "BREAK" STATUS AND CLEARS IT.  GIVES TRAP (OR R1)
/UNLESS EVERYTHING OK.
TTOK,      DAP TTOKX
           JSP TTCKS1
           SZA I
TTOKX,     JMP .
TTCKS2,    SAS (2
           JMP TTERR   /NOT HIS TT
DHANGX,    LAW 10      /BREAK. CLEAR OUT BREAK STATUS
          LSM
           AND I TTFP
           SZA I
           DIP I TTFP  /CLEAR NULL COUNT UNLESS BEING INTERRUPTED
           ESM
           LAW 2       /ERROR CODE FOR BREAK
TTERR,     DAC I (ERCODE) /SAVE ERROR CODE
           LIO 71      /SAVE ADDRESS OF ERROR IN TRAPPC
          DIO I (TRAPPC)
           SZF 6
           JMP R1
           LAW TTTSU
           SPI
           IOR (400000)   /SET OVERFLOW IF PREVIOUSLY ON
          DAC 71
           JMP R0

/IOT TYI.  TYPE IN CHAR TO TOP 6 BITS OF AC.
TYI=.      JSP TTOK
           JSP TTU     /GET CHAR OR HANG
           DAC 70      /STORE CHAR
           JMP TTRETN

/IOT TYO.  TYPE CHAR IN TOP 6 BITS OF AC.
TYO=.      JSP TTOK
           LAW I 7777
           AND 70
           JDA TTS     /TYPE CHAR OR HANG
TTRETN,    SZF 6       /TEST WHETHER 1- OR 2-RETURN IOT

R2,        IDX 71      /NORMAL RETURN 2
R1,        IDX 71      /NORMAL RETURN 1
R0,        LIO DSP1    /NORMAL RETURN 0
R0+1,      LFI         /RESTORE FLAGS
           LAC 70
           LIO 72
           JMP I 71
```

```
/IOT TIS.  TYPE IN STRING TERMINATED BY 74.  AC IS PTR.
TIS,       JSP TTOK
           LAC I (TISMAX
           IDC
           DAC DTEM1
TIS1,      LAC 70         /LOOP.  TEST FOR VALID PTR.
          IDC"U"SCI
           SAD DTEM1
           JMP TIS2       /TISMAX EQUALLED.  WILL BE EXCEEDED.
          AND (177740
           RCL 6S
           SNI"U"SZA I
           JMP TIS3       /TOO LOW OR TOO HIGH
           JSP TTU        /GET CHAR OR HANG
           DCH I 70
           SAD (74
           JMP TTRETN     /TERMINATOR SEEN
           JMP TIS1

TIS2,      LAW 3          /TISMAX ERROR CODE
           JMP TTERR

TIS3,      LAW 4          /ILLEGAL REGION ERROR CODE
           JMP TTERR

/IOT TOS.  TYPE STRING TERMINATED BY 74.  AC IS PTR.
TOS,       JSP TTOK
TOS1,      LIO 70
           LCH I 70       /GET CHAR
           SAD (740000
           JMP TTRETN     /RETURN, LEAVING AC STEPPED.
           DIO 70         /UNSTEP IN CASE OF HANG
           JDA TTS        /TYPE CHAR OR HANG
           LCH I 70       /STEP AC
           JMP TOS1

/SUBROUTINE TO TAKE 6-BIT CHAR OUT OF BUFFER.  HANG USER IF EMPTY.
TTU,       DAP TTSX       /EXIT THROUGH TTS
          ERG
           LAW I 2
           DSC 600
          AND I TTFP
           SAS I TTFP
           JMP TTUFUL     /PF 5 SET.  BUFFER FULL WITH PTRS EQUAL.
          RCR 6S         /SAVE PF 1 IN I.0.
           LAC I TTUP
           SPI I          /TYPEACTIVE?
           SAD I TTSP     /EMPTY?
           JMP TIHANG     /TYPEACTIVE OR BUFFER EMPTY
TTU1,      IDC            /STEP USER PTR
           DAC I TTUP
           JMP TTS5

TTUFUL,    DAC I TTFP     /CLEVERLY CLEARS PF 5
           LAC I TTUP
           JMP TTU1
```

```
/SUBROUTINE TO TYPE 6 BITS IN TOP OF AC.  HANG USER IF BUFFER FULL.
TTS,        0
            DAP TTSX
            LAW I 7777
            SAD TTS
            JMP TTS7    /WARNING CHAR.  GOBBLE IT UP.
            ERG
            LAW 40
            LIO I TTUP
            DSC 600
            IOR I TTFP  /STF 1
            SAS I TTFP  /ALREADY SET?
            JMP TTSBEG  /NOT TYPEACTIVE.  START HIM UP.
TTS1,       LAI
            IDC
            SAD I TTSP
            JMP TOHANG  /BUFFER FULL
            LIO I TTPP
            RIL 8S
            SPI I
            JMP TTS2    /77 NOT SAVED UP
            IDC
            SAD I TTSP
            JMP TOHANG  /NO ROOM FOR 2-BYTE CHAR
            CLC
            XCT TTDCHU  /STORE 77
TTS2,       LAC TTS
TTDCHU,     DCH I .     /STORE CHAR
TTS3,       LAW I 1000
            AND I TTPP
TTS4,       DAC I TTPP  /SET OR CLEAR "SAVED 77" STATUS
TTS5,       SZL I       /SKIP IF EXEC DDT
            IDX I (NOCHAR  /STEP COUNT OF CHARS HANDLED
            JSP ASC6    /RE-ACTIVATE CH 6
TTLCHU,     LCH .       /GET CHAR FOR TTU RETURN
            LRG
TTSX,       JMP .

TTSBEG,     DIO I TTSP  /SET PTRS EQUAL
            AND (-2     /CLF 5
            DAC I TTFP
            AND (211    /KEEP LINK, PF 3, AND PF 6
           SAS (1
            JMP TTS1    /EXPECTING A BREAK: DON'T TCB
            LIO TTNUM
            SSB
            LAW TTTBL"T"10000         /CALCULATE PLACE IN TBL
            IOR TTS
            RAL 6S
            LIO I TTPP
            RIL 8S
            SPI         /"SAVED 77" KEPT IN "1000" BIT
            ADD (100    /77 SAVED UP
```

```
        DAP . 1
        LIO .       /GET CODE
        TCB         /SEND IT
        SAS (TTTBL+115
        SAD (TTTBL+100
        JMP TTS6    /BREAK OR CR-NO-LF.  SET LINK.
        JMP TTS3

TTS6,   LAW 200     /SET LINK
        IOR I TTFP
        DAC I TTFP
        JMP TTS3

TTS7,   LAW 1000    /SET "SAVED 77" STATUS
        IOR I TTPP
        JMP TTS4

/ROUTINES TO HANG USER, ETC.
TIHANG, LIO (TYIHNG
        JMP TTHANG

TOHANG, LIO (TYOHNG
TTHANG; LRG
        IDX TTSX    /IN CASE XDDT
        SZL
        JMP TTSX    /XDDT.  GIVE HIM R2.
GTYBSX, SZF 5
        JMP TTHNG1  /A "DON'T HANG ME" IOT
        JSP HANGS   /HANG USER
        JSP ASC6    /REACTIVATE CHANNEL
        JMP HANG1

/SUBROUTINE TO HANG USER.  HUNG STATUS IN I.O.
HANGS;  DAP HANGSX
        LEM
        LAW I 7777  /LOOK AT TOP 6 BITS OF...
        LSM
        AND I RSTAT /...GUY'S STATUS
        SZA I
        DIO I RSTAT /NO SPECIAL BITS SET; HANG HIM
       ESM         /N.B. GUY'S QUEUE IS IN RUNTM
HANGSX, JMP .

/SUBROUTINE TO RE-ACTIVATE CH 6 AND ISB IF SCANNER STOPPED.
ASC6,  DAP ASC6X
        ASC 600
        CKS
        RIR 2S
        SPI
        ISB 600     /SCANNER STOPPED
ASC6X,  JMP .
```

```
/IOT'S TO LEAVE, ENTER, AND SAVE UP 8-BIT MODE AND CONTROL MODE.
TTMODE,   JSP TTCKS1
          SAD (1
          JMP TTERR    /NOT HIS
          LAC I TTFP
          SZF 3
          DAC 72       /RETURN OLD STATUS IN I.O.
         LAW 4
          SZF I 1      /8-BIT IOT?
          JMP TTMOD1   /NO
          LAC 70       /YES.  SET NEW ALARM CHAR.
         XOR I TTPP
          AND (776000 /ALARM CHAR IN TOP 8 BITS
         XOR I TTPP
          LIO I TTPP
          DAC I TTPP
          SZF 3
          DIO 70       /PUT OLD ALARM CHAR IN AC
          LAW 20
TTMOD1,   DAC DTEM1    /STORE FLAG BIT FOR MODE
         CMA
          LSM
          AND I TTFP
          SZF 2        /ENTER MODE?
          IOR DTEM1    /YES
          DAC I TTFP
TTMODX,   ESM
          JMP TTRETN

/IOT "GET TELETYPE"
GTY,      JSP TTSET
          LIO RSTAT
          LAW 377
          LSM
          AND I TTPP
          SZA I
          DIO I TTPP   /IF NO ONE'S: ASSIGN TO HIM
         ESM
          SZA
          SAD RSTAT
          JMP TTRETN   /NORMAL RETURN
TTHNG1,   CLC          /"DON'T HANG ME" CODING
          DAC 72       /-0 IN I.O. TO INDICATE HANG CONDITION
          JSP ASC6     /REACTIVATE CHANNEL 6
          JMP TTRETN   /GIVE HIM A GOOD RETURN
```

```
/IOT RELEASE TT

RTY,        JSP TTCKS1
            SAD (1)       /NOT YOUR TT?
            JMP TTERR
            DZM HOTFLG
            DZM I TOTFLG
            LAW 7741
            LSM
            AND I TTFP
            DAC I TTFP
            DZM I TTPP
            ESM
            LIO (1)
            LAC TSTAT
            LSM
            SAD BDSTAT
            LAW FSTAT
            DAP RTY1
            LAW I 7777
            AND I RTY1
            SAD (GLPHNG)
            DIO I RTY1
            JMP TTMODX

RTY1,       DCH .
```

```
/IOT TO HANG USER UNTIL BREAK ON TT IN TTNO
DHANG,    DSC 600
          JSP TTCKS1
          LIO (EOTHNG)
          SAS (2)     /BREAK?
          JMP TTHANG  /NO. HANG HIM
          JSP ASC6
          JMP DHANGX  /IN TTOK, CLEAR OUT BREAK AND UNHANG

/EXEC DDT TYO AND TYI
XDDTYO,   JDA XDDTX   /CALLED FROM CORE 10
          CLA
          RCR 6S
          JDA TTS     /TYPE FROM BOTTOM OF IO
          JMP XDDTXR
XDDTXY,   LAI
          LIO DDTFLG
          LFI
          LIA
          JSP HNGDDT
XDDTX1,   CLL"U"CML   /XDDT FLAG FOR TT ROUTINES
          LAC EXECTT
          JDA TTNUM   /SETUP POINTERS
          LIO XDDTEM
XDDTXX,   JMP .

XDDTX,    0           /EXIT ACROSS CORE
          DAP XDDTXX
          DIO XDDTEM
          LIF"U"SCF
          DIO DDTFLG
          JMP XDDTX1

XDDTXR,   LIO DDTFLG  /RETURN ACROSS CORE
          LFI
          LIO XDDTEM
          JMP I XDDTX

XDDTYI,   JDA XDDTX   /CALLED FROM CORE 10
          JSP TTU     /TYPE INTO TOP OF AC
          JMP XDDTXR
          DZM DDTGUY
          JMP XDDTXY

XDDTEM,   0
EXECTT,   0
DTEM1,    0
```

```
/ROUTINE FOR DISPATCHER TO HANG XDDT
HNGDDT,   DAC HDDTX
          LAW I 7777
          LSM
          AND STAT 0  /XDDT IS USER 0
          SWP
          SNI
          DIP STAT 0
          ESM
          JSP ASC6    /RE-ACTIVATE CH 6 AFTER HANGING DDT
HNGDD2,   LIF
          DIO DDTFLG
ESWORG,   LEM
          JMP SWORG   /ON CH 17 BREAK - CALLED FROM XDDT

/ROUTINE TO START UP XDDT WHEN IT COMES INTO CORE
RUNDDT,   LAW I 7777
          LSM
          AND STAT 0
          SZA
          JMP DDTBRK  /BREAK KEY ON EXEC TT
RUNDD1,   ESM
          LAC RCORE
          DAC DCORE
          SAR 2S
          IOR (200000)   /XDDT RUNS IN CORE 10 (RENAMED)
          DCH (ADD .+1)
          RNM
          LAC DDTGUY
          SZA         /DID HE HAVE ANYBODY
          JMP DDTWT1  /YES, GET HIM INTO CORE AGAIN
RUNDD2,   EEM
          LIO DDTFLG
          LFI
          JMP I HDDTX /NO, GO BACK TO DDT'S CORE

HDDTX,    100000
DDTGUY,   0
DDTSU,    100000
DDTNUS,   100001
DDTFLG,   0

DDTBRK,   LAW 1
          DAC STAT 0
          LAC DDTSU
          DAC HDDTX
          DZM DDTGUY
          JMP RUNDD1
```

```
/ROUTINE FOR XDDT TO CALL TO GET A GUY INTO CORE
DDTWNT,   DAC HDDTX
          LEM
          LAW STAT
          AAI
          DAC DDTGUY
DDTWT1,   LAW WHERE-STAT
          ADD DDTGUY
          DAP DDTWT2
          LAW 777
          LSM
DDTWT2,   AND .
          SAD (300      /IS THERE SUCH A GUY
          JMP DDTHLT    /NO
          LAC (300000)
          AND I DDTGUY
          CLI"U"SWP
          SNI
          LAC I DDTGUY
          IOR (10000)
          DIP I DDTGUY    /SET BIT
          ESM
          LAW I 2
          DAC I DCORE /HOLD DDT'S CORE
          JMP HNGDD2

/ROUTINE TO START UP XDDT WHEN HIS USER COMES INTO CORE
DDTWS,    LAC (-10000)    /REMOVE XDDT WANTS BIT
         LSM
          AND I RSTAT
          DIP I RSTAT
          ESM
          LAC RSTAT    /IS THIS GUY DDT CURRENTLY WANTS
         SAS DDTGUY
          JMP SWORG    /NOT GUY: FORGET IT
         LAW STAT 0   /IS GUY: SET DDT RUNNING
         DAC I DCORE
          DAC RSTAT
          JMP RUNDD2

DDTHLT,   ESM
          LAC DDTNUS   /TELL DDT THAT THERE'S NO USER
          DAC HDDTX
          JMP RUNDD2
```

```
/XDDT RPA
DDTRPA,    DAC DDTRPX
DDTRP1,    JSP RDA
           XX
           JMP DDTRDH
           JMP I DDTRPX
DDTRPX,    0

DDTRDH,    LIO (RPAHNG)
           JSP HNGDDT
           JMP DDTRP1

START XX-JMP
```

SWAPPER 10/13/66 =SWAP,56=

```
/PRESWAPPER, HANDLES RUN TIME FOR RUNNING USER

SWAPT,    0             /TIME OF LAST DEBREAK (IN MS.)

SWAP,     DSC 1700      /BREAK ON CHAN 17
         SZS 10
          JSP CEXECTT
          EEM
          RCK
          LAI
          SUB SWAPT     /CALCULATE ELAPSED TIME
         SPA
          ADD (60000.)
          DAC SWAPT     /SAVE ELAPSED TIME
          LIF"U"SCF"U"CLL               /PRESERVE LINK IN IO
         TAD I (TOTRT+1)
          DAC I (TOTRT+1)
          SZL"U"LFI
          IDX I (TOTRT)
          LAC (700000)
          LEM
          AND I RSTAT
          EEM
          SZA
          SAD (400000)
          JMP SWAPH
          DAP I (BITS)
          JMP SWAPI
SWAPH,    LAW 7777
          AND I (BITS)
          ADD SWAPT
          LIA
         AND (37)
          DAP I (BITS)
          XAI
          SZA I
          JMP SWAPI
          LAW I 1
          ADD RUNTM
          LIA
          AND RUNTM
          AND (1777)
          SWP
          SNI I
          JMP SWAPA

          LAW 7400      /GUY CHANGING QUEUE
         AND RUNTM     /IS HE BACKGROUND?
          SZA
          JMP SWAPD     /YES
          LAC RUNTM
          SAL 2S
          SUB (1)
```

```
SWAPG,     DZM HOTFLG
SWAPA,     DAP RUNTM    /UPDATED RUN TIME
SWAPI,     LIO HOTFLG
SWAPC,     SNI
           JMP SWAPB
SWDBK,     LAC RSTAT
           SAS BDSTAT
           SZS I 30
           JMP SWDBK1

SWAPB,     LAC 74
           DAC I (AC)   /SETUP P POINTERS FROM DEBREAK AREA
           LIO 75
           DIO I (PC)
           JSP C16TSV
           LIO 76
           DIO I (IO)
           LIF"U"SCF"U"CLL
           DIO I (FLAGS)
           LEM          /SWAPPER RUNS OUT OF EM
           LAC RUNTM
           DAP I RSTAT
```

```
/SWAPPER PROPER

SWORG,     CLC"U"CLF 7
           DAC HOTFLG   /INIT. HOTFLG
           LAW I 7777   /SEE IF XDDT PERMANENTLY HUNG
          AND STAT 0
           SAS (40000)
           JMP SWLO
           LAW I 2
           SAS I DCORE
           JMP SWLO
           LAW STAT 0
           DAC I DCORE
           DZM DDTGUY
SWLO,      CLA
           SAS LDSTAT
           STF 1
           SAS BDSTAT
           STF 2
SWLOA,     LAW STAT     /INIT TO FIND BEST GUY
           DAP SWV1
           LAW WHERE
           DAP SWV2
           LAC (I)
           DAC SWBQ
SWV1,      LIO .        /GET HIS QUEUE
           SPI          /HUNG?
           JMP SWL1     /YES, IGNORE
           LAW 777      /NO
SWV2,      AND .        /FIND OUT WHERE HE IS
           RAR 6S
           ADD (SWSTB)  /LOOK UP IN SKIP TABLE
          DAP SWV3
SWV3,      XCT .        /SKIP IF ACCESSIBLE
           JMP SWL1     /INACCESSIBLE, IGNORE
          CLA          /FIGURE EFFECTIVE QUEUE          JSP 5
           RCL 6S                                       DAP 2
           SZA                                          JMP 5
           CLI
           RIR 6S
           LAI
           SUB SWBQ
           SMA
           JMP SWL1     /NOT AS GOOD AS BEST SO FAR
          DIO SWBQ     /BETTER, UPDATE BEST SO FAR
           LAC SWV2
           DAP SWV7
SWL1,      IDX SWV1     /LOOK AT NEXT GUY
           IDX SWV2
           SUB (AND)
           SAS WTOP
           JMP SWV1
           LAC SWBQ
           SAD (I)      /DID WE FIND ANYBODY?
           JMP SWBGC    /NO, RUN BEST GUY IN CORE
           CLC
           DAC SWWQ
           LAW CS1
           DAP SWV4
SWV4,      LAC .        /GET CORE STATUS
          SPA
           JMP SWL2     /UNAVAILABLE, IGNORE
          SZA I
```

```
JMP SWL3       /FREE, STOP SEARCHING
```

```
         DAP SWV5
SWV5,    LIO .        /LOOK AT STATUS OF PGM IN THAT CORE
         SPI
         JMP SWL9     /HUNG, EFFECTIVE QUEUE IS (I)
         CLA
         RCL 6S
         SZA
         CLI
         RIR 6S
SWL10,   LAI          /GET STATUS INTO AC TOO
         SUB SWWQ
         SPA
         JMP SWL2     /ALREADY HAVE A WORSE GUY
         DIO SWWQ     /REMEMBER NEW WORSE GUY
         LAC SWV4
         DAP SWT2
SWL2,    IDX SWV4     /LOOK AT NEXT CORE
         SAS (LAC CS3+1)
         JMP SWV4
         LAC SWWQ
         SPA
         JMP SWNN     /DIDN'T FIND ANY CORES AT ALL, WAIT
         SUB SWBQ
         SPQ          /SWAPPABLE PAIR?
         JMP SWBGC    /NO, GO RUN BEST GUY IN CORE
SWL8,    LAW 700      /YES, WHERE IS NON-CORE GUY?
SWV7,    AND .
         CLI
         SZA I
         JMP SWLD     /LITTLE DRUM
         SAD (100)
         STF 4        /FG 4 ON MEANS B.D., OFF MEANS LIMBO
         SZF 3        /FG 3 INDICATES FREE CORE
         JMP SWFC
         LAW WHAT     /INIT. TO SEARCH FOR FREE L.D. FIELD
         DAP SWV6
SWL5,    LAW 777
SWV6,    AND .
         SAD (400)
         JMP SWL4     /FOUND FREE L.D. FIELD
         IDX SWV6
         SAS (AND WHAT+100)
         JMP SWL5
         SZF 4
         JMP SWL11    /GUY IS ON BIG DRUM
         DAP SWV11    /GUY IS IN LIMBO, INIT. TO FIND B.D. FIELD
SWL12,   LAW 777
SWV11,   AND .
         SAD (400)
         JMP SWL11A   /FOUND FREE B.D. FIELD
         IDX SWV11
         SAS (AND WHAT+200)
         JMP SWL12    /KEEP LOOKING
```

SPQ ?! AH! SO!

```
/SORRY, YOU LOSE.
          LIO I SWV7
          LAW I WHERE-STAT
          ADD SWV7
          DAP SWV11A
          LAW 300
          LSM
          DAC I SWV7
          LAC (UNUSED)
SWV11A,   DIP .
          LAW 200
          XAI"U"CLL"U"CML           /WE'LL NEED THE LINK LATER
          SAS (700)
          JMP SWV11C
          EEM
          LAC (DCH SWV11Z)
          JDA 14STXT
          LEM
SSWORG,   ESM
          JMP SWORG

SWV11C,   JDA TTNUM   /SET UP TT POINTERS
          DZM I TTPP  /FLUSH THE PROGRAM NUMBER
          LAW SWV11Y
          DAC SWV11W
SWV11D,   LCH I SWV11W
          SAD (SPO)
          JMP SSWORG
          JDA TTS     /GO TYPE OUT THE CHARACTER
         NOP
          JMP SWV11D

SWV11W,   0

SWV11Z,   472224
          004471
          222265
          647756

SWV11Y,   TEXT /CALL LATER PLEA/
          634577
          157712
          770474
```

```
/MISCELLANEOUS BITS OF CODING
/FOUND FREE CORE
SWL3,    STF 3         /FREE CORE INDICATOR
         LAC SWV4
         DAP SWT2      /REMEMBER WHICH CORE
         JMP SWL8

/SKIP TABLE FOR FIGURING ACCESSIBILITY
SWSTB,   SZF 1         /ON L.D.
         SZF 2         /ON B.D.
         SZF 6         /IN LIMBO
         NOP           /HALTED PGMS ARE NEVER ACCESSIBLE
         NOP           /CORE IS NEVER ACCESSIBLE
         SZF 6         /STARTUP PROGRAM PRIME

/PGM IN CORE IS HUNG
SWL9,    LIO (I)
         JMP SWL10
SWV20,   LIO WHAT      /VARIABLE FOR SWL11 LOOP

/FIND GUY TO PUT ON B.D.
SWL11A,  SZF I 2
         JMP SWL11
         STF 6
         JMP SWL0A
SWL11,   LAC SWV20
         DAP SWV13
         CLC
         DAC SWWQ
SWV13,   LIO .         /SEARCH ONLY LD WE HAVE
         LAW 700
         NAI
         SZA
         JMP SWL14     /NOT A GUY THERE
         LAW STAT
         AAI
SWL13,   DAP SWV12
         SAD (STAT)
         JMP SWL14     /DON'T SWAP XDDT ON FASTRAND
SWV12,   LAC .         /GET GUY'S STATUS
         LIA           /INTO IO
         AND (720000)
         SAS (400000)
         SAD (020000)
         JMP SWL14     /SKIP ANY FASTRAND GUYS
         SZA
         CLI
         SPA
         LIO (I)       /EXCEPT HUNG GUYS
         LAI           /EFFECTIVE Q IN AC,IO
         SUB SWWQ
         SPA
         JMP SWL14     /WORSER GUY ALREADY (SIC)
         DIO SWWQ      /REMEMBER THIS GUY
```

```
          LAW WHERE-STAT
          ADD SWV12
          DAP SWV16
SWL14,    SZF 6         /GET NEXT GUY, CORE OR LD?
          JMP SWL15     /CORE, GO LOOK AT NEXT CORE
          IDX SWV13     /LOOK AT NEXT LD FIELD
          SAD (LIO WHAT+40)
          LAC (LIO WHAT)
          DAP SWV13
          SAS SWV20
          JMP SWV13
          LAW WHAT
          ADD I SWV16
          DAP SWV20
          STF 6         /NO LD FIELDS LEFT, LOOK AT CORES NOW
          LAW CS1
          DAP SWV15
SWV15,    LAC .         /CORE STATUS WORD
          SZM           /IS CORE OK?
          JMP SWL13     /YES, COMPARE GUY WITH OTHERS
SWL15,    IDX SWV15     /GET NEXT CORE
          SAS (LAC CS3+1)
          JMP SWV15
          LAC SWWQ
          SPA
          JMP SWBGC     /NO, WAIT
          SUB SWBQ
          SPQ           /FASTRAND SWAPPABLE PAIR?
          JMP SWBGC
          LAW 400       /YES, WHERE IS HE?
SWV16,    AND .         /WHERE PTR TO GUY FOUND
          SZA
          JMP SWBD      /CORE, GO SET UP BD WRITE
          SAS LDSTAT    /LD, SETUP LD READ UNLESS LD BUSY
          JMP SWBGC     /LD BUSY, WAIT
          LAW WHAT
          ADD I SWV16
          DAP SWV13

SWL17,    LAW I WHAT
          ADD SWV13     /GET L.D. READ FIELD
          DAP SWT3
          RAR 6S        /INTO HI ORDER AC
          SPA           /WHICH L.D.?
          STF 5
          IOR (400000)     /SET READ BIT
          DIP LDIA
          LAW 777
          AND I SWV13 /GET INCOMING PGM
          ADD (STAT)
          DAP LDT2      /SETUP FOR L.D. S.B. ROUTINE
          ADD (WHERE-STAT)
          DAP LDV1
          LAW I CS1-1-400             /GET PHYSICAL CORE
          ADD SWT2
          DAC LDT1
          RAR 4S
          DIP LDCL
          LAC SWT2
          DAP LDV2
```

```
                SZF 3
                JMP SWL19       /READ ONLY, GO DO DIFFERENT THINGS
                LAW 201         /SAVE L.D. READ FIELD FOR NEXT SWAP
                DAC I SWV13
                LIO SWT3        /GET READ FIELD
                SZF 5           /WHICH DRUM
                JMP SWL20
                LAC SWFLDF      /DRUM 0, RESET SAVED L.D. FIELD
                DIO SWFLDF
                JMP SWL21

SWL20,          LAC SWFLDF+1    /DRUM 1, RESET SAVED FIELD
                DIO SWFLDF+1
SWL21,          LIA             /GET WRITE FIELD IN AC AND IO
                RAR 6S
SWL24,          IOR (400000)    /SET WRITE BIT, WRITE ONLY JOINS HERE
                DIP LDWC
                LAW WHERE-STAT
                ADD I SWT2      /GET WHERE PTR. TO GUY BEING WRITTEN
                DAP .+1
                DIO .           /SAY HE'S ON THE DRUM
                LAW WHAT        /GET WHAT PTR. TO WRITE FIELD
                AAI
                DAP SWV17
                LAW I STAT      /SAY DRUM FIELD HAS GUY
                ADD I SWT2
SWV17,          DAC .
SWL23,          CLC             /SET UP STATUS FOR SWAPPING
                DAC LDSTAT
                DAC I SWT2      /SAY CORE IS HELD
                LAW I 2
                DAC LDERC
                SZF 5           /WHICH DRUM?
                JMP SWL22
```

```
        LSM
        DRA
        LAW 48.
        AAI
        DAP LDIA
        DAP LDCL
        LIO LDIA
        DIA
        LIO LDWC
        DWC
        LIO LDCL
        DCL
        ESM
        CLA         /SETUP SKIP FOR SEQ.BRK. ROUTINE
SWL25,  DAP LDV3
        JMP SWORG

SWL22,  LSM         /SAME ROUTINE FOR DRUM 1
       DRA 100
        LAW 48.
        AAI
        DAP LDIA
        DAP LDCL
        LIO LDIA
        DIA 100
        LIO LDWC
        DWC 100
        LIO LDCL
        DCL 100
        ESM
        LAW SPA"U"SMA-SKP
        JMP SWL25

/NORMAL ENTRY TO LITTLE DRUM CODING
SWLD,   LAW WHAT
        ADD I SWV7  /GET READ FIELD
        DAP SWV13
        JMP SWL17   /ENTER L.D. CODING

/READ ONLY CODING
SWL19,  LAW 400
        DAC I SWV13 /MARK READ FIELD AS FREE
        CLA         /SETUP TO NOT WRITE
        DIP LDWC
        JMP SWL23   /GO DO DRUMSWAP

/WRITE ONLY CODING
SWL18,  CLA         /ENTRY POINT. SETUP TO NOT READ
        DIP LDIA
        LAC SWT2    /GET CORE TO SET UP FOR S.B. ROUTINE
       DAP LDV2
        SUB (CS1-1) /GET PHYSICAL CORE
        RAR 4S
        DIP LDCL
        LAC SWV6
        SUB (AND WHAT) /GET WRITE FIELD
        LIA
        RAR 6S
        SPA         /WRITE FIELD DETERMINES DRUM
       STF 5        /DRUM INDICATOR
       JMP SWL24    /ENTER MAINLINE SETUP CODING
```

```
/LITTLE DRUM SEQUENCE BREAK ROUTINE
LD,          DRA 100
LDV3,        SKP            /SKIP IF DRUM 1 IS BEING USED
             DRA            /GET DRA ON CORRECT DRUM
             LAC .          /10 MICROSECS.
             SPI
             JMP LDERR      /ERROR ROUTINE
LDERI,       DZM LDSTAT     /RESTORE STATUS, UNAFFECTED BY EM
             DZM HOTFLG     /MARK ALARM
             ISB 1700       /ACTIVATE SWAPPER
             LAC LDIA
             SMA            /WRITE ONLY?
             JMP LDWO       /YES
             LIO LDT1       /PRE SET-UP CORE OF GUY WHO CAME IN
LDV1,        DIO .          /MARK GUY WHO CAME IN AS BEING IN CORE
LDT2,        LAW .          /STAT PTR. FOR GUY WHO CAME IN
LDV2,        DAC .          /RESTORE CORE STATUS
LDERGI,      LAC 24
             LIO 26
             JMP I 25       /DEBREAK

LDWO,        CLA            /SET CORE TO FREE
             JMP LDV2

LDT1,        0

LDERR,       RIL 1S
             SPI I
             XX             /TIMING ERRORS
             ISP LDERC
             JMP LDERG      /TRY AGAIN
             LAW I STAT
             ADD LDT2
             RCR 9S
             LAW I 3
             EEM
             LSM
             JDA 14SOCT
             LAC (LDERT)
             JDA 14STXT
             LAC LDIA
             XCT LDV3
             AND (-ADD)
             LIA
             LAW I 2
             JDA 14SOCT
             LAC (LDECRT)
             JDA 14STXT
             ESM
             JMP LDERI
LDECRT:      775600
LDERT:       436400
             476151
             005600
```

```
L DERC,     0

L DERG,     CLA
            DIP LDWC
            LAC LDIA
            SMA
            JMP LDWO
            XCT LDV3
            JMP LDERGO
            LIO LDIA
            DIA 100
            LIO LDWC
            DWC 100
            LIO LDCL
            DCL 100
            JMP LDERGI

L DERGO,    LIO LDIA
            DIA
            LIO LDWC
            DWC
            LIO LDCL
            DCL
            JMP LDERGI
```

```
/FOUND FREE L.D. FIELD
SWL4,     CLA            /IS L.D. BUSY?
          SAD LDSTAT
          JMP SWL18      /NO, DO DRUM WRITE TO GET FREE CORE

/THIS SECTION FINDS THE BEST GUY IN CORE TO RUN
SWBGC,    LAW CS1        /INITIALIZE
          DAP SWV8
          LAC (I)        /WORSE THAN WORST QUEUE
          DAC SWBQ
SWV8,     LAC .          /GET CORE STATUS
          SPQ
          JMP SWL6       /NOBODY THERE
          DAP SWV9
SWV9,     LIO .          /LOOK AT STATUS OF GUY THERE
          SPI            /HUNG?
          JMP SWL6       /YES, IGNORE
          CLA            /FIGURE EFFECTIVE QUEUE
          RCL 6S
          SZA
          CLI
          RIR 6S
          LAI
          SUB SWBQ       /COMPARE WITH BEST QUEUE SO FAR
          SMA
          JMP SWL6       /NOT AS GOOD
          DIO SWBQ       /BETTER, REMEMBER THIS GUY
          LAC SWV8       /REMEMBER CORE
          DAP RCORE
          LAC SWV9       /REMEMBER GUY
          DAP RSTAT
SWL6,     IDX SWV8       /LOOK AT NEXT CORE
          SAS (LAC CS3+1)
          JMP SWV8
          LAC SWBQ       /NO MORE CORES, DID WE FIND ANYONE?
          SAD (I)
          JMP SWNN       /NO ONE WANTS TO RUN, WAIT
          DAP SWL28      /SAVE QUEUE OF BEST GUY
          AND (1777)
          SAD (1777)     /X777?
          JMP SWL26      /YES
SWL30,    LAW I CS1-1
          ADD RCORE      /SETUP TOP OF RCORE
          RAR 4S
          DIP RCORE
          SAR 2S         /SETUP TO RENAME
          DCH (ADD SWV10)
SWV10,    RNM
          LAW STAT
          SAD RSTAT
          JMP RUNDDT
          EEM
SWV19,    JMP SWL32      /SWITCH
```

```
SWL31,  LAW SWL32
        DAP SWV19   /RESTORE SWITCH
        LAW 200
        XAI
        LIO C16SU
        SAD (700)
        LIO C16PSU
        DIO I (PC)
        DAC I (BILLTT)
        DZM I (TOTRT)
        DZM I (TOTRT+1)
        DZM I (IOPRT)
        DZM I (IOPRT+1)
        DZM I (BITS)
        DZM I (NOCHAR)
        LSM
        LAC I DAT
        LIO I TME
        ESM
        DAC I (SUPTD)
        DIO I (SUPTD+1)
```

```
SWL32,    LEM
          LAC I RSTAT /GET GUY'S STATUS
          DAP RUNTM
          RAL 5S
          SPA
          JMP DDTWS   /EXEC DDT WANTS TO SEE THIS GUY
          RAR 1S
          SPA
          JMP IOPWS1  /THE IO PROCESSOR WANTS HIM
          RAR 1S
          SPA
          JMP BRKS    /HE HIT THE BREAK KEY
SWL33,    EEM         /HE REALLY WANTS TO RUN
          LIO I (PC)
          DIO 75      /SETUP PC IN DEBREAK AREA
SWL34,    JSP C16TUS  /UNSAVE CORE 16 IF NECESSARY
          LIO I (FLAGS)
          LFI
          LAC I (AC)
          DAC 74
          LIO I (IO)
          DIO 76
          JMP SWDBK   /DEBREAK

SWL26,    LAW STAT    /INIT TO CHANGE ALL X777'S TO X776
          DAP SWV18
SWL28,    LAW .       /LAW X777
SWV18,    SAS .
          JMP SWL29   /NOT X777, IGNORE
          LAW I 1
          ADD I SWV18
          DAP I SWV18 /STORE X776
SWL29,    IDX SWV18
          SUB (SAS STAT-WHERE)
          SAS WTOP
          JMP SWL28
          JMP SWL30
```

```
SWFC,     SZF 4         /WHERE IS BEST GUY?
          JMP SWBDR     /READ FROM FASTRAND
          LAW I WHERE-STAT            /STARTUP STUFF
          ADD SWV7      /GET STAT PTR.
          DAP I SWT2    /SAY HE'S IN CORE
          DAP RSTAT     /SAY HE'S RUNNING
          LAC SWT2
          DAC RCORE     /SAY THIS CORE IS RUNNING
          LIO I SWV7    /GET TT #
          SUB (CS1-1-400)             /GET 400 CORE PTR
          DAC I SWV7    /REPLACE IN WHERE TABLE
          LAW SWL31     /SETUP SWITCH
          DAP SWV19
          JMP SWL30     /GO RUN GUY


SWNN,     SZS 10
          JSP CEXECTT
          LAW 374
          AND TTF
          RAR 2S
          LIA
          LAC I TTF
          RAR 1S
          SPA
          LFI
          CLA
          SAS HOTFLG
          JMP SWNN      /WAIT FOR SOMETHING TO HAPPEN
          JMP SWORG     /SOMETHING HAPPENED, GO HANDLE IT


SWDBK1,   RCK
          DIO SWAPT
          LAC 74
          LIO 76
          ASC 1700
          JMP I 75      /DEBREAK


SWAPD,    LIO I (PRIORITY)
          CLA"U"CMI
          RCR 8S
          JMP SWAPG
```

```
/BREAK KEY HAS BEEN HIT
BRKS,      LAC RSTAT  /GUY BEING FASTRAND SWAPPED?
          SAD BDSTAT
           JMP SWL33   /YES, IGNORE BREAK KEY
           LAC (-40000)   /REMOVE BREAK KEY BIT
          LSM
           AND I RSTAT
           DAC I RSTAT
           ESM
           EEM
           LAC I (BITS)   /IS HE RUNNING UNDER IDDT
          SMA
           JMP BRK1+1  /NO : GO CHECK RESTART MODE
          LAW 77       /WAS THE TT THE BILLTT
           AND I (BILLTT)
           JDA TTNUM   /SET UP POINTERS
          LAW .+3      /USE TT ROUTINES TO CHECK
           DAP TTCKSX
           JMP TTCKS1+2
           SAS (2)      /.+3 : IS IT A BREAK
           JMP BRK1     /NOT BILLTT : IS USER IN RESTART MODE
           LAC I (AC)  /COPY AC,PC,IO,FLAGS
           DAC I (HH16AC)
           LIO I (IO)
           DIO I (HH16IO)
           LIO I (FLAGS)
           DIO I (HH16FG)
           LAC (-500000)  /REMOVE IDDT AND RESTART BITS
           AND I (BITS)
           DAC I (BITS)
           LAC C16BKK
           LIO I (PC)
           DIO I (HH16PC)
           JMP BRK3

BRK1,      LAC I (BITS)   /IS USER IN RESTART MODE
BRK1+1,    AND (-100000)
           SAD I (BITS)
           JMP ESWORG  /USER NOT IN RESTART MODE
           IOR (40000) /SET DEBREAK LEGAL BIT
           DAC I (BITS)
           LIO I (PC)
           DIO I (RESTPC)
           LAW RESTSU
BRK3,      DAC 75      /RESET PC TO APPROPRIATE POSITION
          LAW 1        /SET TO HIGH QUE
           DAC RUNTM
           JMP SWL34
```

```
/FASTRAND SWAPPER
SWBD,    SZF I 4      /SWAP OR WRITE ONLY
        JMP SWBDW    /WRITE ONLY
         LAW WHAT
         LIO SWFBDF   /GET SLOT TO WRITE ON (WS)
        AAI
         DAP SWBDV1
         LAC I SWV7   /GET SLOT TO READ FROM (RS)
        DAC SWFBDF   /NEW SWAPPING SLOT
        ADD (WHAT)
         DAP SWBDV2
         LAC I SWV16 /GET 40C WHERE C IS CORE INVOLVED
        DAC I SWV7   /SAY GUY ON RS IS NOW IN CORE
        ADD (CS1-1-400)
         DAP SWV4     /CS PTR TO CORE
        DIO I SWV16 /SAY GUY WHO'S IN CORE IS NOW ON WS
        LAW I STAT
         ADD I SWV4
SWBDV1,  DAC .        /WHAT PTR TO WS
SWBDL1,  LAW STAT     /READ ONLY COMES IN HERE
SWBDV2,  ADD .        /WHAT PTR TO RS
         DAC I SWV4   /SAY CORE HAS GUY WHO USED TO BE ON RS
         LAW 201      /CODE: SAVED FOR SWAPPING
        SZF 3
         LAW 400      /READ ONLY, CODE: FREE
         DAC I SWBDV2    /SETUP NEW STATUS OF RS
SWBDL2,  LAW I WHAT 100 /WRITE ONLY COMES IN HERE
        ADD SWBDV1   /CALCULATE WRITE SLOT
        RCR 9S
         LAW I WHAT 100
         ADD SWBDV2   /CALCULATE READ SLOT
         RCR 9S
         LAW I CS1-1
         ADD SWV4     /GET CORE
         RAR 6S       /POSITION
         DCH (ADD .+1)   /SETUP TO RENAME THAT CORE TO 0
        RNM
         EEM
         LAC I (AC)
         DAC I (FASTAC)
         LAC I (PC)
         DAC I (FASTPC)
         DIO I (AC)
         LAC C16FSW   /STARTING ADDRESS FOR SWAP
         SZF 3
         LAC C16FIN   /SA FOR READ ONLY
         SZF I 4
         LAC C16FOU   /SA FOR WRITE ONLY
         DAC I (PC)
         LEM
```

```
            LAW SWBDL5
            SZF I 4
            LAW SWBDL4  /SPECIAL CODING ON RETURN FROM WRITE ONLY
            DAP SWBDV4
            LAC I SWV4
            LSM
            DAC BDSTAT
            LAC I BDSTAT
            DAC FSTAT
            ESM
            LAW 1
            DAC I BDSTAT
            JMP SWORG

/WRITE ONLY ENTRY
SWBDW,      LAW CS1-1-400
            ADD I SWV16
            DAP SWV4    /SETUP CS PTR TO CORE
            LAC SWV11   /GET FREE BD FIELD FOUND
            DAP SWBDV1  /SETUP PTR
            JMP SWBDL2

/READ ONLY ENTRY
SWBDR,      LAW WHAT
            ADD I SWV7  /GET RS PTR
            DAP SWBDV2
            LAW I CS1-1-400
            ADD SWV4    /GET CORE PTR
            DAP I SWV7
            JMP SWBDL1
```

```
/ERROR RETURN FROM FASTRAND SWAP
SWFERR,   LAW I STAT
          ADD RSTAT
          RCR 9S
          LAW I 3
          LSM
          JDA 14SOCT
          LAC (SWFERT)
          JDA 14STXT
          ESM
/RETURN FROM FASTRAND SWAPPING (CHANNEL 16)
SWFAST,   LAC I (FASTAC)
          DAC 70
          LAC I (FASTPC)
          DAC 71
          LEM
          LSM
          LAC FSTAT
          DAC I RSTAT
          DAP RUNTM
          DZM BDSTAT
          ESM
SWBDV4,   JMP .       /SWITCH

SWBDL4,   DZM I RCORE /WRITE ONLY, MARK CORE AS FREE
          LAW I STAT
          ADD RSTAT
          DAC I SWBDV1   /MARK WS AS CONTAINING GUY
         ADD (WHERE)
          DAP SWBDV5
          LAW I WHAT
          ADD SWBDV1
SWBDV5,   DAP .       /MARK GUY AS BEING ON FASTRAND
SWBDL5,   DZM HOTFLG  /TELL SWAPPER IN ALL 3 CASES
         ISB 1700
          JMP R0

SWFERT:   662226
          006451
         517756
```

```
/SUBROUTINES TO SAVE AND UNSAVE CORE 16 VARIABLES IF NECESSARY
C16TSV,  DAP C16TSX
         LAW I 7777
         NAI
         RIL 2S      /PC IN IO
        SPI I
C16TSX,  JMP .
         DAC C16TS1
         LAW C16TEM
         DAC C16TS2
C16TSI,  LAW I 10
         DAC C16TS3
C16TSL,  LAC I C16TS1
         DAC I C16TS2
         IDX C16TS1
         IDX C16TS2
         ISP C16TS3
         JMP C16TSL
         JMP C16TSX

C16TUS,  DAP C16TSX
         LAW I 7777
         NAI
         RIL 2S
         SPI I
        JMP C16TSX
         DAC C16TS2
         LAW C16TEM
         DAC C16TS1
         JMP C16TSI

C16TS1,  0
C16TS2,  0
C16TS3,  0
```

```
IOPWS1,   LAC (-20000)   /SET HUNG BIT @ CLEAR WANTS BIT
         LSM
         AND I RSTAT
         DAC I RSTAT /AND FLUSH WANT BIT
         ESM
          LAC RSTAT
          DAC ISTAT
          LAW I 1
          DAC I RCORE
          LAC RCORE
          DAC ICORE
          SAR 2S
          IOR (100000)   /RENAME CORE TO 4
          DCH (ADD .+1)
          RNM
          LAW I 7777
          AND ICORE   /RETURN TO IOP WITH REALCORE IN AC
         EEM
          JMP I IOPWGO   /GO TO CORE 17

CEXECTT,  DAP CEXECX
          LAW TTP"T"LAC
          IOR EXECTT
          SAL 2S
          DAP CEXEC1
          ADD (3)
          DAP CEXEC2
          LAW 101
CEXEC1,   DAC .
CEXEC2,   DZM .
          LAT
          AND (77)
          DAC EXECTT
          SAL 2S
          ADD (TTP)
          DAP CEXEC3
          ADD (3)
          DAP CEXEC4
          LAW 105
CEXEC3,   DAC .
          LAW STAT 0
CEXEC4,   DAC .
CEXECX,   JMP .
```

```
LDIA,     0
LDWC,     0
LDCL,     0
LDSTAT,   0
BDSTAT,   0
FSTAT,    0
SWWQ,     0
SWBQ,     0
SWT1,     0
SWT2,     0
SWT3,     0
RUNTM,    0
HOTFLG,   -0
WTOP,     WHERE+1

RCORE,    0
ICORE,    0
DCORE,    0
RSTAT,    0
ISTAT,    0
TSTAT,    STAT+1

WHERE,    402
          REPEAT 177, 300              /UNUSED
WHAT,     201            /L.D. FIELD 0 SAVED FOR SWAPPING
          REPEAT 37, 400 /OTHER FIELDS EMPTY
          201            /FIELD 40
          REPEAT 37, 202 /SECOND DRUM HASN'T ARRIVED
         201             /B.D. SLOT 0
          REPEAT 61, 400 /49. SLOTS
          REPEAT 16, 202 /14. MEANINGLESS PLACES

CS1,      0
CS2,      STAT 0         /XDDT IS IN CORE 10
CS3,      -3             /CORE 14 HASN7T ARRIVED YET
SWFLDF,   0              /L.D. FIELD 0 IS SWAPPING FIELD
         40
SWFBDF,   100
```

```
VARIABLES
CON, CONSTANTS
FOO,

7500/
DSPTB,    140000 ELGIOT
          140000 DHANG
          140000 TYI
          140000 TYO
          140000 TIS
          140000 TOS
          140000 GTY
          140000 RTY
          140000 TTCKS
          140000 TTMODE
          140000 RSMC
          REPEAT 2, 140000 ELGIOT
          140000 PSU
          140000 DELAY
          REPEAT 13, 140000 ELGIOT
          140000 DWPP
          REPEAT 4, 140000 ELGIOT
          140000 DPEEK
          140000 ELGIOT
          140000 DCLR /GRDR GET THE READER
          140000 DRPA /RPA  READ PUNCHED TAPE ALPHABETIC
          140000 DRLSRD  /RRDR RELEASE THE READER
          140000 DGETPU  /GPUN GET PUNCH
          140000 DPPA /PPA  PUNCH PUNCHED TAPE ALPHABETIC
          140000 DRLSPU  /RPUN RELEASE PUNCH
          140000 .RCK
          REPEAT 30, 140000 ELGIOT
DSPTB+100,
```

```
C16TB,      REPEAT 100, 360000 10
C16TB+100,

C16BKP,     360000 10
C16BKK,     360000 10
C16HH,      360000 10
C16HLT,     360000 10
C16SU,      360000 10
C16PSU,     360000 10
C16FIN,     360000 10
C16FOU,     360000 10
C16FSW,     360000 10

7714/       /CORE 15 ENTRIES
1SECJA,     DCH 1SECA
1SECJB,     DCH 1SECB
SORADR,     DCH SORX
1MIADR,     DCH 1MIX
15SOCT,     0
15SOC1,     0
15STXT,     0
15STX1,     0
DAT,        0
TME,        0
TOTFLG,     0

7740/      /CORE 17 ROUTINES
DATADR,     DCH DATX
CONADR,     DCH CONX
IOPADR,     DCH IOPX
IOPWGO,     JDA .

EXPUNGE TYIHNG,TYOHNG,GTYHNG,EOTHNG,RPAHNG,UNUSED,CORHNG,CLKHNG,GLPHNG

START HLT-JMP
```

CORE 10 ENTRIES TO CORE 14 11-10-66 (14TO10,10)

```
100002/   DCH STAT
          DCH DDTWNT
          DCH XDDTYO
          DCH XDDTYI
          DCH DDTRPA
          DCH XRDIN
          DCH RDRLS

100015/   DCH WTOP
          DCH WHERE

START HLT-JMP
```

CORE 15 ENTRIES TO CORE 14 (14TO15,14)

```
150010/  DCH CONX
         DCH OCTR
         DCH 1MIX
         DCH RSTAT
         -STAT
         DCH R0
         DCH R1
         DCH R2
         DCH ELGIOT
         DCH DSP1
         DCH HOTFLG
         DCH C16RH
         DCH FSTAT
         DCH BDSTAT
         STAT 1
         STAT 2
         DCH SWFAST
         DCH SWFERR
```

START HLT-JMP

CORE 17 ENTRIES TO CORE 14 (14TO17,10)

```
177700/   DCH DATX
          DCH IOPX
          DCH R0
          DCH R1
          DCH R2
          DCH ESWORG
          DCH HOTFLG
          DCH ISTAT
          DCH RSTAT
          DCH ICORE
          -STAT
          DCH INIT
          DSWAPX
          DCH DCORE

START CAL 400
```

```
XDDT PART 1  11-10-66  (XDDTA,13)

ASC=720051
CAC=720053
CBS=720056
CKS=720033
DSC=720050
EEM=724074
ESM=720055
IOH=730000                  /IN-OUT HALU
ISB=720052
LAT=762200
LEM=720074
MSM=720054
PPA=720005
PPB=720006
RPA=720001
RPB=720002
RRB=720030
SFT=660000
SZS=640000
TYI=720004
TYO=720003
 XX=760400

DBA=722061
DCL=720063
DIA=720061
DRA=722062
DWC=720062
ERM=720065
LRM=720064
RCC=723022
RCH=722022
RCK=720032
RNM=720066
RRC=722122
RRI=720037
RRO=720017
RSC=721122
RSM=720067
RTB=720035
SSB=724122
TCB=724022
TCC=725022
```

```
HALT=HLT
TYIHNG=IOT 100
SUPPC=40  /PMACE TO PLANT INSTRUCTION
PC=35
DDTCOR=100000

0/          JMP RLSE
            JMP USERR
STAT,       0
DWANT,      0
DDUO,       0
DDUI,       0
DDTRPA,     0
DDTCLR,     0
DDTRRD,     DDTCOR MSE
IOPDNE,     0             /CORE 17 LINK INTO CORE 10
12,         JMP ER4       /(IT'S 20 NOW)
DCONT,      0
14,         0             /UNUSED
WTOP,       0
WHERE,      0
17,         0             /UNUSED

20,         CLA           /HALT PROCEDURE
            MSM
            SAS I DCONT
            JMP ER5       /IOP BUSY
            JSP I IOPDNE
            HLT           /MAY PUSH CONTINUE
26,         JMP T+1       /ENTRY FROM RESTART PROCEDURE

40/
LOWMIM,     DDTCOR+4200/

DEFINE      DISP MC,UC
            Z=LC-LSE 44
            Z"T"1000 UC-LSE 44
TERMINATE DISP

DEFINE      LETTER A,B
            DISP A+MSE-44,B
TERMINATE LETTER

DEFINE      SQOZE A,B,C
            A"T"1600.+B"T"40.+C
TERMINATE SQOZE

L=0         R=1

DDM=(DDTCOR)
EIC=(DDTCOR+SGP)
EVC=(DDTCOR+EST)

C20=(20)
C77=(77)
CI=(I)
CJ=(JMP)
C4=(400000)
```

```
LOW,      0              -0 /INITIAL SYMBOL TABLE
          0              DDTCORE
          SQOZE  0, 27, 46          MSK
          SQOZE  0, 36, 46          EST
          SQOZE  0, 15, 46          MEM
          0                 400000

/OPERATE GROUP (IN EVENT TIME ORDER)

          SQOZE 15, 15, 23          CCI
          SQOZE 26, 13, 32          LAP
          SQOZE 26, 13, 36          LAT
          SQOZE 15, 26, 15          CLC
          SQOZE 15, 26, 13          CLA
          SQOZE 15, 26, 23          CLI
          SQOZE 26, 13, 32          OPR 100      /MAP-CLA OPR
          SQOZE 26, 13, 36          OPR 2000     /LAT-CLA OPR
          SQOZE 15, 27, 13          CMA
          SQOZE 15, 27, 23          CMI
          SQOZE 22, 26, 36          HLT
          SQOZE 35, 41, 32          SWP
          SQOZE 26, 13, 23          LAI
          SQOZE 26, 23, 13          LIA
          SQOZE 35, 36, 20          STF
          SQOZE 15, 26, 20          CLF
NOPCOD,   SQOZE 30, 31, 32          NOP
          SQOZE 31, 32, 34          OPR

/SPECIAL OPERATE GROUP (IN EVENT TIME ORDER)

          SQOZE 26, 20, 23          LFI
          SQOZE 26, 23, 20          LIF
          SQOZE 35, 44, 26          SZL
          SQOZE 35, 15, 23          SCI
          SQOZE 35, 15, 20          SCG
          SQOZE 15, 26, 26          CLM
          SQOZE 23, 20, 23          IFI
          SQOZE 23, 23, 20          IIF
          SQOZE 35, 15, 27          SCM
          SQOZE 15, 27, 26          CML
          SQOZE 23, 16, 13          IDA
          SQOZE 23, 16, 15          IDC
          SQOZE 13, 13, 23          AAI
          SQOZE 23, 13, 23          IAI
          SQOZE 42, 13, 23          XAI
          SQOZE 35, 32, 31          SPO
```

```
/SKIP GROUP

                SQOZE 15, 26, 31            CMO
                SQOZE 35, 32, 33            SPQ
                SQOZE 35, 44, 27            SZM
                SQOZE 35, 30, 23            SNI
                SQOZE 35, 32, 23            SPI
                SQOZE 35, 44, 31            SZO
                SQOZE 35, 27, 13            SMA
                SQOZE 35, 32, 13            SPA
                SQOZE 35, 44, 13            SZA
                SQOZE 35, 44, 20            SZF
                SQOZE 35, 44, 35            SZS
                SQOZE 35, 25, 32            SKP

/STANDARD IOSTRUCUIONS

                SQOZE  0,  0, 23            I
                SQOZE 13, 16, 16            ADD
                SQOZE 13, 30, 16            AND
                SQOZE 13, 35, 15            ASC
                SQOZE 15, 13, 15            CAC
                SQOZE 15, 13, 26            CAL
                SQOZE 15, 14, 35            CBS
                SQOZE 15, 25, 35            CKS
                SQOZE 16, 13, 15            DAC
                SQOZE 16, 13, 32            DAP
                SQOZE 16, 14, 13            DBA
                SQOZE 16, 15, 22            DCH
                SQOZE 16, 15, 26            DCM
                SQOZE 16, 23, 13            DIA
                SQOZE 16, 23, 31            DIO
                SQOZE 16, 23, 32            DIP
                SQOZE 16, 23, 40            DIV
                SQOZE 16, 34, 13            DRA
                SQOZE 16, 35, 15            DSC
                SQOZE 16, 41, 15            DWC
                SQOZE 16, 44, 27            DZM
                SQOZE 17, 17, 27            EEM
                SQOZE 17, 34, 27            ERM
                SQOZE 17, 35, 27            ESM
                SQOZE 23, 16, 42            IDX
                SQOZE 23, 31, 22            IOH
                SQOZE 23, 31, 34            IOR
                SQOZE 23, 31, 36            IOT
                SQOZE 23, 35, 14            ISB
                SQOZE 23, 35, 32            ISP
                SQOZE 24, 16, 13            JDA
                SQOZE 24, 27, 32            JMP
                SQOZE 24, 35, 32            JSP
                SQOZE 26, 13, 15            LAC
                SQOZE 26, 13, 41            LAW
                SQOZE 26, 15, 22            LCH
                SQOZE 26, 17, 27            LEM
                SQOZE 26, 23, 31            LIO
                SQOZE 26, 34, 27            LRM
```

```
        SQOZE 26, 35, 27          LSM
        SQOZE 27, 37, 26          MUL
        SQOZE 32, 32, 13          PPA
        SQO[E 32, 32, 14          PPB
        SQOZE 34, 13, 26          RAL
        SQOZE 34, 13, 34          RAR
        SQOZE 34, 15, 15          RCC
        SQOZE 34, 15, 22          RCH
        SQOZE 34, 15, 25          RCK
        SQOZE 34, 15, 26          RCL
        SQOZE 34, 15, 34          RCR
        SQOZE 34, 23, 26          RLI
        SQOZE 34, 23, 34          RIR
        SQOZE 34, 30, 27          RNM
        SQOZE 34, 32, 13          RPA
        SQOZE 34, 32, 14          RPB
        SQOZE 34, 34, 14          RRB
        SQOZE 34, 34, 15          RRC
        SQOZE 34, 34, 3:3         RRI
        SQOZE 34, 34, 31          RRO
        SQOZE 34, 35, 15          RSC
        SQOZE 34, 35, 27          RSM
        SQOZE 34, 36, 14          RTB
        SQOZE 35, 13, 16          SAD
        SQOZE 35, 13, 26          SAM
        SQOZE 35, 13, 34          SAR
        SQOZE 35, 13, 35          SAS
        SQOZE 35, 15, 26          SCL
        SQOZE 35, 15, 34          SCR
        SQOZE 35, 20, 36          SFT
        SQOZE 35, 23, 26          SIL
        SQOZE 35, 23, 34          SIR
        SQOZE 35, 35, 14          SSB
        SQOZE 35, 37, 14          SUB
        SQOZE 36, 13, 16          UAD
        SQOZE 36, 15, 14          TCB
        SQOZE 36, 15, 15          TCC
        SQOZE 36, 43, 23          TYI
        SQOZE 36, 43, 31          TYO
        SQOZE 42, 15, 36          XCT
        SQOZE 42, 31, 34          XOR
SGP,    SQOZE  0, 42, 42          XX
        SQOZE  0,  2, 35          1S
        SQOZE  0,  3, 35          2S
        SQOZE  0,  4, 35          3S
        SQOZE  0,  5, 35          4S
        SQOZE  0,  6, 35          5S
        SQOZE  0,  7, 35          6S
        SQOZE  0, 10, 35          7S
        SQO[E  0, 11, 35          8S
        SQO[E  0, 12, 35          9S

EST,    DDUCORE LOW
```

```
RLSE,       JSP I DDTRRD
LSE,        JSP LCC        /MISTEN LOOP
LSE+1,      CLC"U"CLF 7
            DAC SP2
            DAC OPO
LSS,        CMC
            DAC CHI
            DAC SP1
LSP,        DZM WRD
SSN-1,      LAC CUN        /(IOR)
SSN,        DIP SGN
BSLASH,     DZM DOM
            DZM SYL
N2,         DZM SYM L      /LOOP FOR NEXT SYLLABLE
            DZM SYM R
            DZM FSM
            CLC"U"CLF 4
            DAC LET
            DAC CC
LSR,        LIO SK1        /LOOP PER CHAR.  CHEAP PLACE...
            DIO WEA        /...TO DO INITIALIZATION.
            LAW VF5+1
            DAP VF5
            LAW LWU
            DAP BAX
            JSP I DDTI     / TYPE-IN, HANG IF OO CHARS
            RAL 6S
            DAC CH
CDB,        LAW DTB        /USED AS DTB
            DAC FM1        /NON-ZERO
CAD,        ADD CH         /USED AS ADD
            DAP /+1
LISL,       MIO .
            ISP CAS
            JMP UPPER
LSU,        SZF I 4        /"QUOTE" MODE?
            JMP NOTQM      /NO
            IDX CC
            SAS (4
            JMP LN4
QUOTE2,     LAC FSM
            JMP N1
```

```
/NO-EVAL ROUTINES

ULC,      LAW I 2      /77 SEEN
          DAC CAS
          JMP MSR

EOT,      CLC          /EOT SEEN. RESET EVERYTHING FOR LARRY
          DAC MSK
          DAP UL
          DZM LL
          LAC (DCH
          DAC MEM
          DAC TAS
          LAW 8.
          DAC RADIX
          DZM USER
          DZM NUM
          LAW PI
          DAP PNS
          LAW PEV
          DAP PA1
          JMP LSE+1

CHH,      LIO LET      /NUMBER-SIGN LOOKUP
          LAC CHI
          SPI"U"SMA
          JMP SHALL
          LAW CHTBL
SEEK,     DAP .+1
/AND
YE,       LAC .
/SHALL    SZA
/FIND     JMP CHH1
ERR,      LAW CHARAC R?  /PRINT Q-MARK.  IGNORE LAST CHAR.
          JDA TYS
          JMP LSR

XCL,      LAW XCTBL    /EXCLAMATION MARK
          JMP SEEK

DOT,      LIO CC       /PERIOD
          LAC LOC
          SPI I
          LAC DNM
          DAC SYL
          DAC DNM
          LAW 44
          DAC T2
          JMP LN1

QUO,      LAC FSM      /" MEANS TAKE AS INTERNAL
          SZA
          JMP N1
          STF 4
          JMP LSR
```

```
Q,          LAC LWT      /"Q" IS LAST QUANTITY TYPED
            JMP N1

DAQ,        LAW 7777     /<DEFINES SYM AS ADDRESS OF Q
            AND LWT
            JMP COM+1

COM,        LAC LOC      /COMMA DEFINES SYM AS LOC
COM+1,      DAC DF1
            DZM FL1
DEF,        LAC LET      /DEFINE SYMBOL
SK1,        SZA
            JMP ERR
            JSP DE
            JMP PN2

DEL,        LAW 3        /#,DELETE
            JDA TYS
DEL1,       JMP PN2      / END OF NO EVAL ROUTINES

/SET OUTPUT RADIX

RADX,       SNI"U"SZM    /SET RADIX
            SAD (1
            JMP ERR
            DAC RADIX
            JMP LSE

/ARITHMETIC FUNCTIONS

PLS,        LAC CAD      /PLUS SIGN
            JMP SSN

MIN,        SPI          /MINUS SIGN
            DIO WRD
            MAC CSU      /(SUB)
            JMP SSN

ISC,        LAC CAN      /(AND)  LOGICAL "AND".
            JMP SSN

UNI,        JMP SSN-1    /LOGICAL "OR"
```

```
/PRINT "Q" IN FORM SPECIFIED

EQL,        DAC LWT     /PRINT INTEGER IN CURRENT RADIX
            JSP LCT
            JDA OPT
PN2,        JSP LCT
            JMP LSS

DEC,        DAC LWT     /PRINT AS DECIMAL INTEGER
            LAC RADIX
            DAC T
            LAW 10.
            DAC RADIX
            JSP LCT
            JEA OPT
            LAC T
            DAC RADIX
            KMP PN2

ARW,        DAC LWT     /PRINT AS IOSTRUCTION
            JSP MCT
            DAC PI      /JDA PI, BUT DON'T SET FLAG 2
            LAW PN2
            STF 3       /FORCE SYMBOLIC PRINTOUT
            JMP PI+2

PBX,        DAC LWT     /PRINT AS INTERNAL
            JSP LCU
            JDA TYSA
            JMP PN2

SQP,        DAC LWT     /PRINT AS SQOZE CODE
            JSP LCT
            DAC SYM R
            DZM SYM L
            LAW PN2
            DAP SPX
            JMP SPTMIM

/SET CURRENT CORE

CORE,       SNI I
            JMP ERR
            RAR 6S
            AND (170000
            DIP MEM
            DIP TAS     /YES,VIRGINIA, IT'S NECESSARY
            DZM LOC
            JMP MSE
```

```
/REGISTER EXAMINATION AND CHANGE

VBR,        AND (177777 / :
            DIP TAS
            SPI
            JMP VBR1
            DIP MEM
            JMP TA5

VBR1,       AND (170000
            SZA
            BRING1      /CALL BRING WITHOUT CHANGING MEM
            JMP VBR1X

BAC,        LAW OPT     /OPEN BRACKET (BAR-CONSTANT)
            JMP BAS+1

BAI,        LAW TYSA    /CONTROL "I"  (BAR-INTERNAL)
            JMP BAS+1

BAS,        LAW PI      /CLOSED BRACKET (BAR-SYMBOLIC)
BAS+1,      DAP BAX
VBR1X,      MAC OPT
BAR,        SPI
            JMP TA6
            LIO SP1
            DIO SP2
            SPI
            JMP TA5
            AND (177777)
TR3,        DAC SP2
SP5,        JSP LCT
            LAC I SP2
            JMP TA7

CR,         DAC LWT     /LINE FEED
            JDA MRF
            LAW 7715
            JDA TYS
            JMP MSE+1

UC8,        JDA MRF     /> MEANS MAKE CORR. AND OPEN REGISTER
            JMP TA6
```

```
BS,         JDA MRF      /BACKSPACE
            LIO SP2
            SPI I
            JMP BS2
            IDX LOC
BS+5,       DAC LWT
            JMP TA3+2

FS,         JDA MRF      /ARROW UP (FORWARD SPACE)
            JSP LCC
            LAW I 1
            LIO SP2
            SPI I
            JMP FS1
            ADD MOC
            JMP BS+5

TAB,        JDA MRF      /MEMORY FIELD SWITCH
TA3,        DAC MWT
            JSP MCC
TA3+2,      JDA PAD
            LAW CHARAC R/
            JDA TYS
TA5,        DIM LOC
            DAP LOC
            LIO MEM
            EIO TAS
TA6,        MIO C4
            DIO SP2
            DAP TAS
            JSP LCT
            BRING
            MAC I UAS
TA7,        EAC LWT
BAX,        JDA .        /PI, OPT OR MWT
            DZM OPN
            JMP PN2
```

```
/SYMBOL ROUTINES

VAL,      DAC DF1       /OPEN PAREN, SETS UP VALUE FOR DEFIOITION
          JMP LSS

TBL,      JSP I DDTCLR    /READ SYMBOL PUNCH
          JMP RDNY
          DZM FL1
          JSP SOI
MR1,      JSP GWD
          DAC SYM L
          JSP GWD
          AND (177777
          DAC SYM R
          IOR SYM L
          SZA
          JMP TBM1
TBN,      JSP LCT
          LAC EST
          JDA OPT
TBM,      KSP RBK       /SKIPS REST OF TAPE
          JMP TBM

KIL,      SPI           /KILL SYMBOL(S)
          JMP KI3
KI2,      LIO LEU
          SPI
          JMP ERR
KI1,      LAC I EV2     /USED AS LAC I
          IOR KI1
          DIP I EV2
          JMP LSE
```

```
/ZERO CORE

PUL,        SPI             /LOWER LIMIT SETUP
            JMP ERR
            DAC FA
            JMP LSS

ZRO,        JSP OK          /ZERO CORE TO CONTENTS OF M#+3
            LAW 7777
            SPI
            DAC WRE
            DIP WRD
            AND FA
            SPI
            CLA
            IOR MEM
            DAC T
            LIO NUM
            BRING
ZR2,        SUB MEM
            SUB WRD
            SZM
            JMP LSE
            DIO I T
            IDX T
            JMP ZR2
```

```
/SEARCHES

EAS,        LAW EA1      /EFFECTIVE ADDRESS SEARCH
            JMP WS

NWS,        LAC SK2      /NOT WORD SEARCH
            DAC WEA
WDS,        LAW WS1      /WORD SEARCH
WS,         SPI
            JMP ERR
            DAP WS2
            JSP MCC
            LAC USER
            DAC MWSU
            DZM MWSXU
MWSNU,      DZM MWSFUI
            SZA I
            SAS MEM
            KMP OOJ MWS
            LIO MWSXU    /MOBY WORD SEARCH:  IF USER=MEM=0, SEARCH...
            IDX USER     /...ALL USERS FOR WRD/
            SAD (200
            JMP LOOK1    /RESTORE USER; EXIT TO LSE.
            ADD STAT
            DAC EAS1
            LAW I 7777
            AND I EAS1
            SAD (760000
            JMP WS3+5
NOTMWS,     BRING
            LAC LM
            IOR MEM
            DAC T
            DIP T2
WS4,        DZM SYM
            DAP T3
            LAC I T2
WS2,        JMP .        /EA1 OR WS1
```

```
/ROUTINES COOCERNED WITH TIME SHARING

LOOK,     SPI           /SET USER BEIOG EXAMINED
          JMP CREATE    /GO CREATE NEW USER
          LIA
          SUB (82.      /WOULD BE 200, BUT TOP OF...
                        /STAT + WHERE TBMS SOMETIMES PATCH AREA
          SMA
          JMP ERR
          LAC STAT
          AAI
          DAC T
          LAW I 7777
          AND I T
          SAD (760000
          JMP USERR
MOOK1,    DIO USER
          JMP MSE

HOLD,     SPI I
          JMP HALT1     /HALT USER
          LAW (IOR (LAC))              /PREVENT USER FROM HALTING
          JMP HOLD1

FREE,     MAW (AND (-LAC))             /ALMOW USER TO HALT
HOLD1,    DAP HOLD2     /SET OR CMEAR HELD BY XDDT BIT
          JSP NOUU0
          SPI I
          JMP ERR       /WORKS FOR CURRENT USER ONLY
          BRING1
          LAC I C4      /(400000)
HOLD2,    XCT .
          DAC I C4
          JMP LSE
```

```
SETPTR, SPI I        /FREE TT PTRS
        SPA
        JMP ERR
        SAL 2S
        ADD (141000
        DAC T
        SUB (141000+100"T"4)        /100 TT'S
        SMA
        JMP ERR
        MAW 70"T"4
        AND T
        SZA I        /SET UP TT BUFFER PTRS
        LAW I 100    /0-7←100
        SAS (10"T"4)
        SAD (20"T"4)
        LAW I 300    /10-27←400
        SMA
        LAW I 1100   /30-77←1400
        CMA
        LIO OPT
        SIL 3S
        ADD (140000
        AAI
        LIA
        LAW 101
        LSM
        DAC I T
        IDX T
        DIO I T
        IDX T
        DIO I T
        IDX T
        LAC I T
        DZM I T
        ESM
        JDA OPT
        JMP LSE
```

```
BGN,        BRING1        /START USER
            SPI
            JMP BGN3      /NO ARG
            AND (170000
            SZA I
            JMP BGN1      /CORE 0
            SZS I 60
            JMP ER4
BGN1,       JSP NOTU0
            LIO OPT
BGN2,       DIO I (PC     /ENTRY FROM "STRTUP"
BGN3,       LAC STAT
            ADD USER
            DAC T
            LAW 1
            DAP I T       /HIGH QUEUE
            JMP LSE

XEC,        SNI I 60      /EXECUTE ARG
            JMP ER4       /SS6 NOT UP
            DAC .+1
T,          0
            NOP
            JMP LSE

/READ OR VERIFY PAPER TAPE

SVFY,       LAW VF2       /SPECIAL VERIFY
            DAP VF5
VFY,        SPI I         /VERIFY CORE AGAINST BINARY TAPE
            JMP ERR
            JSP I DDTCLR
            JMP RDNY
            JSP LCC
            LAC RB2       /(SAD I)
            JMP RD1
```

```
REPEAT 1IF VP .-LSE+44-1000, PRINT .MYSTIC 1000 EXCEEDED.

RD,         SPI I         /READ BINARY TAPE INTO CORE
            JMP ERR
            JSP OK
            JSP I DDTCLR
            JMP RDNY
            LAC BS1       /(DAC I)
RD1,        DIP VF4
            JSP SOI
            BRING
VF1,        LAC MEM
            DIP T
            LAC T
            SUB LL
            SUB MEM
            SPA
            JMP VF2
            ADD LL
            SUB UL
            SZM
            JMP VF2
            LAC  I LA
VF4,        T             /DAC I OR SAD I
            JMP VF2
            SZA I
            JMP VF5       /SPECIAL VERIFY, IGNORE ZERO ON TAPE
            LAW I 7777
            AND I T
            ADD T
            SUB MEM
            XOR I LA
            SZA I         /SPECIAL VERIFY, IGNORE "." IN ADDRESS PART
VF5,        JMP .         /JMP .+1  OR  JMP VF2
            JSP PAC
            JSP LCT
            LAC I LA
            JDA LWT
            JSP LCC
VF2,        IDX T
            IDX LA
            SAD RB1
            JSP RBK
            JMP VF1
```

```
/BRANCHES.  PLACED HERE TO SAVE ROOM IN DISPATCH AREA.

/REST OF LISTEN LOOP

NOTQM,   CLA
         RCL 9S
         DAC T2
         SUB (44
         SPA
         JMP LN
         ADD TLS       /(JMP LSE)
         DAP LSX
         SUB (JMP DEL1  /LAST NO-EVAL ROUTINE
         SPQ
         JMP LSX
         LAW SYL
         LIO LET
         SPI I         /SKIP IF LETTER NOT SEEN
         JSP EVL
         JMP EV4
         LAW CHARAC RU  /IGNORE INPUT AND START OVER
ER2,     JDA TYS
         JMP LSS

EV4,     DAP SGN
         LAC WRD
SGN,     XX            /OPERATOR AND SYLLABLE ADDR.
         DAC WRD
         LIO CHI
         SPI
         LAC LWT
         DAC OPT
LSX,     JMP .         /I. O. MINUS IF NO ARG.  ARG IN AC.

UPPER,   JSP TYPOUT
         XCT LSL
         RIL 9S
         JMP LSU

LN,      ADD (44-12    /LETTER-NUMBER LOGIC
         SPA
         JMP N
         DZM LET
LN1,     DZM CHI
         IDX T2
         IDX CC
         SAS (4
         JMP LN3
         LIO SYM R
         DIO SYM L
         DZM SYM R
```

```
LN3,        SUB (6
            SZM
            JMP LSR
            LAC SYM R
            RAL 2S
            ADD SYM R
            RAL 3S
            ADD T2
            DAC SYM R
            LAW I 4
            ADD CC
            SMA
            JMP LSR
LN4,        LAC FSM      /ENTRY FROM QUOTE
            RAL 6S
            ADD CH
            DAC FSM
            LAW 77
            SAD CH
            JMP ULC
            LAW 3
            SZF 4
            SAS CC
            JMP LSR
            JMP QUOTE2

N,          LAC SYL
            RAL 3S
CUN,        IOR T2       /USED AS IOR
            DAC SYL
            LAC DNM
            RAL 2S
            ADD DNM
            RAL 1S
            ADD T2
            DAC DNM
            JMP LN1
```

```
/REST OF NUMBER-SIGN LOOKUP

SHALL,      LAC SYL
            SPA
            JMP ERR
            SAL 2S
            SUB (100"T"4)
            SMA
            JMP ERR
            ADD (100"T"4+141000
            JMP DDS

CHH1,       SAD SYM R
            JMP FIND
            IDX YE
            IDX YE
            JMP YE

FIND,       IDX YE
            DAP .+1
            JMP .

STATUS,     LAC STAT
            ADD USER
            JMP DDS
C,          LAW MEM
            JMP F+1

M,          LAW MSK
            JMP F+1

F,          LAW EST
F+1,        IOR DDM
DDS,        DAC SP1
N1,         DZM CHI
            DAC SYL
            DAC DNM
            JMP N2

/MODE CHANGING ROUTINES

LOT,        LAW TYSA
            JMP CNS+1

SMB,        LAW PI
            JMP CNS+1

CNS,        LAW OPT       /SYMBOLIC-CONSTANT-FLEXO SWITCH SETUP
CNS+1,      DAP PNS
            JMP LSE

OAD,        LAW PVM
            JMP RAD+1

RAD,        LAW PEV       /OCTAL-RELATIVE SWITCH SETUP
RAD+1,      DAP PA1
TLS,        JMP LSE       /USED AS LSE
```

```
/REST OF BS

BS2,      IDX SP2
SP3,      DAC LWT
          DIP MEM
          JDA PAD
          LAW CHARAC R/
          JDA TYS
          LAC TAS
          DIP MEM
          KMP SP5

/REST OF FS

FS1,      ADD SP2
          DAP SP2
          JMP SP3

/REST OF KIL

KI3,      LAC (DDTCOR LOW
          DAC EST
          JMP LSE

/REST OF TBL

TBL1,     JSP GWD
          DAC DF1
          LAC SYM L
          LIO SYM L
          RIL 1S
          SMA"U"SPI
          JMP MR1
          AND (177777
          DAC SYM L
          JSP DE
          JMP MR1

/REST OF EFFECTIVE ADDRESS SEARCH

EA1,      DAC EAS1    /SAVE INSTRUCTION
          LIA
          AND (770000
          SAD (CAL
          JMP EA4
          SAD (JDA
          JMP EA6
          SAD (LAW I
          JMP WS3
          AND (760000
          SAS (OPR    /FLUSH THESE
          SAD (SPO
          JMP WS3
```

```
          SAS (SKP
          SAD (SFT
          JMP WS3
          SAD (IOT
          JMP WS3
          SAS (DCH
          SAD (LCH
          JMP EA2
          LAI
EA3.,     AND CI          /IS THERE AN I BIT
SK1.,     SZA
          JMP EA2         /YES
          LAC EAS1
          AND (760000
          SAD (XCT)
          JMP EA11
          SAS (LCH
          SAD (DCH
          JMP EA5
WS8,      LAW 7777        /NO, AND OFF INSTRUCTION PART OF FIRST WORD
          AND I T2

WS1,      XOR WRD         /COMPARE
CAN,      AND MSK         /USED AS AND
WEA,      XX              /SZA OR SZA I
          JMP WS3         /SETUP NEXT WORD

WS6,      LAC MWSU
          SAD MEM
          SAS MWSFTI
          JMP NOTMW1
          IDX MWSFTI
          JSP LCC
          LAC USER
          DAC MWSXU
          JDA OPT
          LAC (FLEXO "L"
          JDA TYS
          JSP LCC
NOTMW1,   LAW LCC

PAC,      DAP PAX         /HIT, PRINT OUT
          LAC T
          DAP LOC
          JDA PAD
          LAW CHARAC R/
          JDA TYS
          JSP LCT
          LAC I T
          JDA LWT
PAX,      JSP .
```

```
WS3,        IDX T         /INDEX
            SUB MEM
            SUB UL
            SPQ
            JMP WSNW
WS3+5,      LAC MWSU
            SAS MEM
            JMP LSE
            JMP MWSNU

WSNW,       LAC T
            JMP WS4

EA4,        LAC WRD       /CAL FINDS 100,101, AND ITS ADDRESS
            SAS (100
            SAD (101
            JMP WS6
            JMP WS8       /CHECK ADDRESS

EA6,        JSP EAS1+1    /JDA,CHECK THING ADDRESSED
            IDX EAS1      /CHECK ADDRESS+1
EA7,        AND (7777
            JMP WS1

EA2,        LAW EA3.      /GET REG. REFERENCED FOR I,LCH,DCH
            JMP EA12

EA11,       LAW EA1
EA12,       DAP EA13
            LAI
            JDA EAS1
            IDX SYM
            SAD (10.
            JMP WS3
            LAI
            DAP T2
            LAC I T2
            LIA
EA13,       JMP .

EA5,        LAI           /LCH,DCH PTR
            LIO EAS1      /INSTR.
            RIL 5S
            SPI
            IDC
            JDA EAS1      /CHECK LOC. REFERENCED BY POINTER
            LAC T2
            JMP EA7       /CHECK LOC OF POINTER

EAS1,       0             /SPECIAL COMPARE
EAS1+1,     DAP EAS2
            LAW 7777
            AND EAS1
            XOR WRD
            AND MSK
            SZA
EAS2,       JMP .         /MISS,RETURN
            JMP WS6       /HIT,PRINT AND QUIT
```

```
/REST OF "LOOK"; CREATE NEW USER.

CREATE,    LIO (500
           LAC (140000
           LSM
           IOR I WTOP
           DAC T
           SUB WHERE
           DAC OPT
           SUB (82.    /TOP OF STAT + WHERE TBLS PATCH AREA
           SMA
           JMP ER5
           IDX I WTOP
           DIO I T
           ESM
           LAC OPT
           DAC USER
           JDA OPT
           BRIOG1
           LAW 11
           DAC T
           MIO NUM
STRTU1,    DIO I T
           IDX T
           SAS CI
           JMP STRTU1
           LAC (TYIHOG
STRUUP,    BRING1
           DCM I C4    /CLEAR "BITS" IN CASE HALTIOG
           DAC I (SUPPC
           LIO (SUPPC
           JMP BGN2

/REST OF "HOLD"; HALT CURRENT USER.

HALT1,     SAS (1
           JMP ERR
           JSP NOTU0
           LAC (HALT
           JMP STRTUP


START
```

```
XDDT PART 2  11-10-66  (XDDTB,11)

/SUBROUTIOES

MRF,        0              /PUT AC IN CURRENTLY OPEN REG...
            DAP MRX        /...IF THERE IS ONE.
            LAC OPN
            SMA
            SPI
            JMP MRF1
            MAC SP2
            LIO MRF
            SMA
            JMP SPECIAL
            JSP OK
            BRING
            DIO I TAS
MRF1,       LAC MRF
MRX,        JMP .

SPECIAL,    SAD (DDTCOR+MEM
            JMP ER4        /TRIED TO CHANGE C#
            DIO I SP2
            JMP MRF1

/SYMBOLS

DE,         DAP DEX        /DEFINE SYMBOL
            JSP EVL
            JMP DF2
DE1,        LAC (DDTCOR+LOWLIM+6)
            SUB EST
            SMA
            JMP DEX        /NOP TO READ SYMS BELOW LOWLIM
            IDX EST
            LIO FM1
            LAC MEM
            SNI I
            LAC C4
            DAC TYS
            LIO DF1
            SAD I EST
            JMP DE2
            LAW I 2
            ADD EST
            DAC EST
DE2,        DIO I EST
            LAW I 1
            ADD EST
            DAC EST
            LIO SYM R
            DIO I EST
            SUB ONE
            DAC EST
            LAC SYM L
            SZA I
            JMP DE3
            IOR C4
            DAC I EST
            LAW I 1
```

```
DE3,    ADD EST
        DAC PAD
        SUB ONE
        DAC EST
        DZM I EST
        LAC TYS
BS1,    DAC I PAD     /USED AS DAC I
        JMP DEX

DF2,    LAC DF1
        SAD I ES4
DEX,    JMP .
        DAC I ES4     /VALUE IS CHANGING
        LAC EV2
        SUB (DDTCOR+LOW
        SPA
        JMP DEX
        LAC I EV2     /KILL INITIAL SYMBOL
        IOR KI1       /(LAC I)
        DIP I EV2
        JMP DE1       /DEFINE SYMBOL

EVL,    DAP EVX       /EVALUATE SYMBOL
        LAC EST
        DAC ES4
EVG,    DAC EV2
        LAC I ES4
        SPA
        JMP ESN
        SZA I
        JMP EV5
        RAL 1S
        SPA
        JMP ESI
        CLA
ES3,    SAS SYM L
        JMP ESI
        LAC SYM R
RB2,    SAD I ES4     /USED AS SAD I
        JMP EV3
ESI,    IDX ES4
EV6,    IDX ES4
        SAS EVC
        JMP EVG
EV8,    IDX EVX
EV3,    IDX ES4
EVX,    JMP .

ESN,    IDX ES4
        LAC I EV2
        RAL 1S
        SPA
        JMP ESI
        SAR 1S
        JMP ES3
```

```
EV5,     IDX ES4
         LAC I ES4
         SMA
         SAD MEM
         JMP EV6
         IDX ES4
EV7,     LAC I ES4
         SZA I
         JMP EV5
         SPA
         IDX ES4
         IDX ES4
         IDX ES4
         SAS EVC
         JMP EV7
         JMP EV8

PI,      XX              /PRINT INSTRUCUION
         STF 2
PI+2,    DAP PX+1
         JSP PEV
         LAC PI
         SUB CI
         SPA
         JMP PPK
         DAC PI
PR1,     LAW Ø
         JDA TOU
         MAW CHARAC RI
         JDA TYS
PPK,     CLA
         JDA TOU
         XCT EA+1
         JMP PVL
         LAC LWT
         AND (760000
         SAD (SFT
         JMP I66
         SAD PR1         /LAW Ø
         JMP PLO
         RAR 1S
         SZA
CSU,     SUB (320000 /USED AS SUB
         SPA
         JMP PLO
PVL,     LAC PI
PVL+1,   SZA I
         SZF 1 I
PV3,     JDA OPT
PX,      CLF 7           /EXIT
PX+1,    JMP .

PLO,     JSP PEV
         JMP PA1+1
```

```
I66,      LAW 1          /1S-9S
          ADD PI
          AND PI
          SZA
          JMP PVL
          LAW PA1+1
          DAP PEX
          LAC EIC
          JMP EAK+2

PAD,      0              /PRINT ADDRESS
          DAP PX+1
          LAW 7777
          AND PAD
          DAC PI
          CLF 1
PA1,      JSP PEV        /PEV OR PVL
PA1+1,    LAW CHARAC R+
          JDA TYS
          JMP PVL

PEV,      DAP PEX        /SYMBOL LOOKUP SUBR
          LAW I 7777
          AND PI
          SAD (OPR       /DETECT OPERATES
          JMP I76
          AND (760000
          SAS (SKP
          SAD (SPO
          JMP SEV
EAK,      DAP EA+1
          LAC EST
EAK+2,    DAC ES4
          CLF 1
EAL,      LAC I ES4
          LIO I ES4
          SZA I
          JMP EI1
          RAL 1S
          SPA
          JMP EI3
          LAC ES4
          SPI
          IDX ES4
          DAC OP1
          SPI I
          CLI
          DIO T3
          IDX ES4
```

```
EA,         LAC I ES4
EA+1,       SKP I
            JMP SKO
            XOR PI
            SPA
            JMP EIX
            LAC PI
            SUB I ES4
            SPA
            JMP EIX
            SZF I 1
            JMP PSW
            LAC I ES4
            SUB I OP2
            SZM
            JMP PSW
EIX,        IDX ES4     /LOOK AT NEXT SYMBOL
            SAD EVC     /END OF TABLE
            JMP EIX+5
            SAS EIC
            JMP EAL
EIX+5,      XCT EA+1
            JMP PEX
            SZF I 1
            JMP PVL     /TYPE OUT REST IN OCTAL
            LAC PI
            SUB I OP2
            LIA
            SZA         /DETECT NEG NUMS
            JMP I77
EIY-1,      DIO PI
EIY,        JSP SPT     /PRINT SYMBOL
            LAC PI
SK2,        SZA I
            JMP PX
            XCT EA+1
            JMP .+2
PEX,        JMP .
            CMA
            DAC T22     /MASK
            JMP EIX

EI1,        IDX ES4     /STUFF FOR SYMBOL SEARCH
            LAC I ES4
            SMA
            SAD MEM
            JMP EIX
            IDX ES4
EI2,        LAC I ES4
            SZA I
            JMP EI1
            SPA
            IDX ES4
            IDX ES4
            IDX ES4
            SAS EIC
            JMP EI2
            JMP EIX+3
```

```
EI3,        SPI
            IDX ES4
            IDX ES4
            JMP EIX

I76,        DAC T1
            LAC PI
            SAS (NOP
            JMP SEV+2
            CLA
            DAP EA+1
            LAC (DDTCORE+NOPCOD
            JMP EAK+2

SEV,        DAC T1        /SAVE INSTRUCTION
            LAC PI
SEV+2,      CMA
            DAC T22       /MASK
            LAW 600       /SPA"U"SMA-SKP
            JMP EAK

SKO,        IOR T1
            SAS I ES4
            JMP EIX
            SZF 1
            XOR T1
            SZA I
            JMP EIX
            XOR PI
            LIA
            AND T22
            SZA
            JMP EIX
            DIO PI
            LAC (FLEXO "U"
            SZF 1
            JDA TYS
PSW,        LAC I OP1
            DAC SYM R
            LIO T3
            DIO SYM L
            LAC ES4
            DAC OP2
            STF 1
            XCT EA+1
            JMP EIY
            JMP EIX

I77,        LAW I 7777
            AND PI
            SAS (770000
            JMP EIY-1
            LAW CHARAC R-
            JDA TYS
            LAC PI
            CMA
            JMP PV3
```

```
SPT,      DAP SPX     /SYMBOL PRINT SUBROUTINE
          LAW 7777
          AND I OP2
          SAS I OP2
          JMP SPTMIM  /INST.
          ADD PI
          SZF I 2
          JMP SPT1
          SAD LOC
          JMP PRNTPT
SPT1,     LIA
          SUB I OP2
          SUB (100    /LIMIT TAGS TO (VALUE)+100
          SMA
          SZF 3
          JMP SPTLIM
          LAI
          JMP PVL+1   /PRINT AS OCTAL

SPTLIM,   LAC (DDTCORE SYM L
          DAC T3
SPB,      LAW SPD
          DAP SPJ
SPN,      DZM OP1
SPR,      IDX OP1
SPV,      LAC I T3
          AND (177777
SPJ,      SUB .
          SPA
          JMP SPP
SPU,      DAC I T3
          JMP SPR

PRNTPT,   DZM PI
          DIM OP2
          LAW CHARAC R.
          JDA TYS
          JMP SPX

SPP,      LAC OP1
          SCR 1S
          SZA I
          JMP SPS
          ADD SPT     /(SPX)
          DAP .+1
          LAC .
          SPI I
          RAR 6S
          JDA TOU
SPS,      IDX SPJ
          SAS (SUB SPD+3
          JMP SPN
          IDX T3
          SAS (DDTCORE SYM+2
          JMP SPB
SPX,      JMP .
```

```
/UNSQUOZE TABLE.  MUST FOLLOW SPX.

SPL,      FLEXO  01   FLEXO  23
          FLEXO  45   FLEXO  67
          FLEXO  89   FLEXO  AB
          FMEXO  CD   FLEXO  EF
          FLEXO  GH   FLEXO  IJ
          FLEXO  KL   FLEXO  MN
          FLEXO  OP   FLEXO  QR
          FLEXO  ST   FLEXO  UV
          FLEXO  WX   FLEXO  YZ
          1603        /FLEXO .#

SPD,      3100
          50
ONE,      1

/TYPEOUT SUBROUTINES

MCC,      DAP LC1     /LOWER-CASE-CARRIAGE-RETURN
          LAW 76      /TYPE CRLF
          JDA TYS
MC1,      JMP .

LCT,      DAP LCX     /LOWER-CASE-TAB
          CLA
          JDA TYSA
MCX,      JMP .

LWT,      0           /LAST WORD TYPED
          DAP PNX
          LAC LWT
PNS,      JDA PI      /PI, OPT, OR TYSA
PNX,      JMP .

TYS,      0           /TYPE SYMBOL, ETC.
          DZM TSWICH
TYS+2,    DAP TYX
          LAW I 3
          DAC TYSVAR
TYL,      LAC TYS
          RAL 6S
          DAC TYS
          AND C77
          DAC CH      /SAVE CHAR
          SAD TSWICH  /IF 0, DON'T TYPE SPACES
          JMP TYL1
          SAD C77     /IF 77, TYPE "CHAR"
          JMP TYSA1
          JDA TOU
TYL1,     ISP TYSVAR
          JMP TYL
          LAC LWT
TYX,      JMP .
```

```
TYSA,       0               /TYPE 3 CHARS
            DAC TSWICH
            LIO TYSA
            DIO TYS
            JMP TYS+2

TYSA1,      LAC TYS
            RAL 6S
            DAC TYS
            AND C77
            DAC CH
            ISP TYSVAR      /TYPOUT UNLESS 77 IS THIRD BYTE
            JSP TYPOUT
            SAS CH
            JMP TYL1
            RAL 6S
            IOR C77
            JMP TYPOU2

TYPOUT,     DAP TYPEX       /TYPE CONTROL CHARACTERS
            LAC CH
            SAS (12
            SAD (15
            JMP TYPEX
            SAD (4
            JMP TYPEX
            IOR (40
            SAD CH
            JMP TYPEX
            RAL 6S
            IOR (FLEXO " ")
            JDA TOU
            RAR 6S
TYPOU2,     JDA TOU
            RAR 6S
            JDA TOU
TYPEX,      JMP .           /AC HAS CH IF NOTHING HAPPENED
```

```
OPT,        0               /ANY RADIX PRINT
            DAP OPX
            DZM OP1
OPA-1,      LAC OPT
OPA,        DAC OP2
            CLI"U"SWP
            RCL 1S
            DIV RADIX
OP1,        0
            SAS OP1
            JMP OPA
            SWP
            ADD (20
            JDA TOU
            LAC OP2
            DAC OP1
            SAS OPT
            JMP OPA-1
            LAW 10.
            SAS RADIX
OPX,        JMP .
            LAW CHARAC R.
            JDA TYS
            LAC OPT         /IS THIS NEEDED?
            JMP OPX

TOU,        0               /TYPE CHAR FROM BOTTOM OF AC
            DAP TOUX
            LIO TOU
            JSP I DDTO      /TYPE FROM BOTTOM OF I. O.
            LAC TOU
TOUX,       JMP .

/PAPER TAPE SUBROUTINES

SO1,        JDA RDB
SOI,        JDA RDB         / SKIP OVER INPUT ROUTINE
            SPI I
            JMP SO1
RBK,        DAP RBX         /READ A BLOCK INTO BUFFER
            JSP RDB+1
RB3,        LAW BUF
            DAP RB1
            DAP LA
            LAC DDM
            DIP LA
            DZM CHI         /CKSUM
            DIO T2
            DIO T
            SPI
            JMP RLSE
            JSP RDB+1
            DIO CH
            LAW I 1
            ADD CH
            SUB T2
            AND (-77
            SZA
            JMP RBX+1
```

```
RBØ,        JSP RDB+1
            DIO I RB1
            LAC I RB1
            ADD CHI
            DAC CHI
            IDX RB1
            IDX T2
            SAS CH
            JMP RBØ
            ADD CHI
            ADD T
            JDA RDB
            SAD RD4       /WORD JUST READ IS IN RD4
RBX,        JMP .
RBX+1,      LAC (FLEXO SUM /CHECKSUM ERROR
ER3,        JDA TYS
            JMP RLSE

GWD,        DAP GWX       /GET WORD FROM READER BUFFER
GWD+1,      LAC LA
            SAS RB1
            JMP GWD1
            JSP RBK
            JMP GWD+1
GWD1,       DAP GWD2
            IDX LA
GWD2,       LAC .
GWX,        JMP .

RDB,        Ø             / RPB SUBROUTINE
RDB+1,      DAP RDX
            LAW I 3
            DAC RD6
RD3,        JSP I DDTRPA
            RIR 8S
            SPI I
            JMP RD3
            RIL 2S
            LAC RD4
            RCL 6S
            DAC RD4
            ISP RD6
            JMP RD3
            LIO RD4
            LAC RDB
RDX,        JMP .

/TIME SHARING SUBROUTINES

OK,         DAP OKX       /SS6 PROTECT EXEC AND USER ZERO
            CLA
            SAD MEM
            JSP NOTUØ
            SAS MEM
            SZS 6Ø
OKX,        JMP .
ER5,        ESM
ER4,        LAW CHARAC R?
            JMP ER3
```

```
NOTUØ,     DAP NOTUØX
           CLA
           SAS USER
NOTUØX,    JMP .
           JMP ER4

BRING=JDA .
WANTED,    Ø            /GET USER INTO CORE ZERO
           DAP WANTX
           LAC MEM
           SZA
           JMP WANTX-1
WANT1,     DIO SAVEIO
           LIO USER
           SNI I        /Ø"L" FOR NO BRIOG
           JSP I DWANT / USER WANTED IN CORE ZERO
           LIO SAVEIO
WANTX-1,   LAC WANTED
WANTX,     JMP .

BRING1=JDA .
WANT2,     Ø            /BRING REGARDLESS OF VALUE IN MEM
           DAP WANTX
           LAC WANT2
           DAC WANTED
           JMP WANT1

USERR,     LAW FLEXO  NO
           JDA TYS
           CLA
           JDA TOU
           LAW FLEXO  US
           JDA TYS
           LAW FLEXO  ER
           JMP ER3

RDNY,      LAW FLEXO  BU
           JDA TYS
           LAW FLEXO  SY
           JMP ER3

CHTBL,     IRP [X,,T,M,S,C],[Y,,F,M,STATUS,C]
           CHARAC R'X-26
           JMP Y
           ENDIRP
           Ø

XCTBL,     IRP [X,,C,A,R,S,I],[Y,,CNS,OAD,RAD,SMB,LOT]
           CHARAC R'X-26
           JMP Y
           ENDIRP
           Ø
```

```
DTB,      DISP PLS,LSE
          DISP XCL,ISC
          DISP QUO,SVFY
          DISP CHH,CORE
          DISP ERR,EOT
          DISP ERR,EAS
          DISP ISC,FREE
          DISP ERR,BGN
          DISP VAL,HOLD
          DISP DEF,BAI
          DISP DEC,CR
          DISP PLS,KIL
          DISP COM,LOOK
          DISP MIN,ERR
          DISP DOT,NWS
          DISP BAR,ERR
          LETTER 0,SETPTR
          LETTER 1,Q
          LETTER 2,RADX
          LETTER 3,SQP
          LETTER 4,TBL
          LETTER 5,UNI
          LETTER 6,VFY
          LETTER 7,WDS
          METTER 8,XEC
          LETTER 9,RD
          DISP VBR,ZRO
          DISP PUL,ERR
          DISP DAQ,ERR
          DISP EQL,ERR
          DISP UC8,ERR
          DISP PBX,ERR
DTB 40,   DISP ERR,ERR
          REPEAT 3,LETTER .-DTB-27,ERR
          LETTER .-DTB-27,BSLASH
          LETTER .-DTB-27,ERR
          LETTER .-DTB-27,FS
          LETTER .-DTB-27,ARW
          REPEAT 23,LETTER .-DTB-27,ERR
          DISP BAC,ERR
          DISP TAB,DEL
          DISP BAS,ERR
          DISP BS,ERR
          DISP ULC,ERR
DTB+100,
```

```
/"PERMANENT" VARIABLES

MSK,      -0          /FIRST 4 STAY TOGETHER.  MASK FOR WS.
LL,       0           /LOWER LIMIT FOR WS AND RD
UL,       7777        /UPPER LIMIT FOR WS AND RD
NUM,      0           /VALUE STORED BY ZRO

USER,     1
RADIX,    8.
FA,       0           /FIRST ADR FOR "Z".  SET AT PUL BY SEMICOMON.
MEM,      DDTCOR      /CURRENT CORE
LOC,      0           /CURRENT LOCATION

/OTHERS

WRD,      0           /INPUT WORD FOR LSE
SYL,      0           /OCTAL VERSION OF INPUT SYLLABLE FOR LSE
DNM,      0           /DECIMAL   DITTO
FSM,      0           /FLEXO   DITTO
CAS,      0           /CASE FOR LSE.  SET TO -2 WHEN 77 CH SEEN.
CC,       0           /NUMBER OF CHARS THIS SYLLABLE FOR LSE
CHI,      0           /+0 IF ANY CHARS THIS WORD FOR LSE; ...
                      /...-0 IF NONE.  ALSO TEM (CKSUM) FOR RBK.
LET,      0           /+0 IF LETTER THIS SYL FOR LSE; -0 IF OOT.
CH,       0           /CHAR FOR LSE, TYS AND TYSA, AND TYPOUT...
                      /...TEM (END TEST) FOR RBK.
SYM,      0           /L HALF OF SYMBOL FOR LSE, TBL, DE, PEV,...
                      /...SPT, EVL.  TEM (DEPTH OF TRACE) FOR WS.
SYM+1,    0           /R HALF
SP1,      0           /SET IN LSE.  -0 IF NO #-SIGN THIS WORD...
                      /...VALUE IF #-SIGN SEEN.
SP2,      0           /SET BY LSE AND REGISTER EXAMINING ROUTINES...
                      /...ADR OF OPEN #-SIGN REG.  MINUS IF NONE.
OPN,      0           /0 IF REG OPEN; -0 IF NOT...
                      /...SET AT LSE AND BAX.
TAS,      DDTCOR      /ADR OF OPEN REG.  SET BY REG EXAMINING...
                      /...ROUTIOES.  USED BY MRF IN CLOSING REG.
```

```
FL1,      0             /ZERO IF SYM TO BE DEFINED BY DE AS LOCAL;...
                        /...NON-ZERO, GLOBAL.  SET AT LSE, COM, TBL/
DF1,      0             /VALUE OF SYM TO BE DEFINED BY DE...
                        /...SET AT COM, VAL, TBL.
EV2,      0             /SET BY EVL.  POINTS AT FISST WORD OF...
                        /...SYMBOL.  USED BY KIL.
ES4,      0             /SET BY EVL.  POINTS AT VAL OF SYM.  USED...
                        /...BY DE.  AMSO SYM VAL PTR FOR PEV.
T1,       0             /TEM (INSTR) FOR PEV
T22,      0             /TEM (MASK) FOR PEV
U3,       0             /TEM FOR PEV AND SPT
OP2,      0             /TEM (VALUE) FOR PEV AND SPT.  TEM FOR OPT.
RB1,      DDTCOR+BUF    /READER BUFFER INPUT PTR...
                        /...USED BY VFY, RBK, GWD/
LA,       0             /READER BUFFER OUTPUT PTR
RD4,      0             /BINARY WORD BEIOG BUILT BY RDB
RD6,      0             /ISP COUNTER FOR RDB
TYSVAR,   0             /COUNTER FOR TYS AND TYSA
TSWICH,   0             /FLAG FOR TYS AND TYSA...
                        /...IF ZERO, DON'T TYPE SPACES.
T2,       0             /TEM FOR LSE, WS, RBK
SAVEIO,   0             /SAVED I. O. FOR BRING
MWSU,     0             /USER ON ENTRY TO MOBY WORD SEARCH
MWSXU,    0             /USER ON EXIT FROM MOBY WORD SEARCH
MWSFTI,   0             /FIRST TIME INDICATOR FOR MOBY WS

CON,      CONSTANUS

BUF,      .+100/

FOO,      FLEXO FOO

REPEAT 0IF P,[
PRINT /FOO+1/
-[RD-MSE+44-1000]/      PRINT /SPACE LEFT IN DISPATCH AREA/
]

REPEAT 1IF P,EXPUNGE L,R

START HLT-JMP
```